

6

Work, Power and Energy

Ye ek bahut hi joridar, easy and scoring chapter hai jisse Mains or Advance dono mein sawal puche jaate hai isko bahut acche se karna bcz COM, rotation, electrostatic etc mein yah bahut use hoga.



WORK

Work done (dW) by force $\vec{F}$ is given by

$$\star dW = \vec{F} \cdot d\vec{s}$$

$$dW = \vec{F} \cdot d\vec{s}$$

displacement of point of application of force

Chota sa $\vec{F}$ WD jo $\vec{F}$ Ne kia

Vo force hai jiska WD hum nikal rahe hai

$$\star (WD)_{\text{by Force } F} = W_{\text{net}} = \int dw = \int \vec{F} \cdot d\vec{s}$$

$$\star \text{ If } F \text{ is const} \Rightarrow (WD)_{\text{by } F} = \vec{F} \cdot \int d\vec{s} = \vec{F} \cdot \vec{S}$$

$$WD = \vec{F} \cdot \vec{S} \text{ or } WD = \vec{F} \cdot \vec{d}$$

Here $\vec{S}$ or $\vec{d}$ are displacement.

Constant force $\vec{F}$ ka WD $\vec{F} \cdot \vec{d}$ hota hai ye kabhi mat bhulna bcz WD by gravity, any other external constant force etc mein yah bahut use hoga. Constant force ka matlab hai magnitude ke saath direction bhi constant hona dhyhan rakhna iss baat ko.



WORK DONE BY CONSTANT FORCE $\vec{F}$

$$WD = \vec{F} \cdot \vec{d}$$

$$WD = F d \cos \theta$$

angle between $\vec{F}$ & $\vec{d}$

WD by force $\vec{F}$

Magnitude of $\vec{F}$

magnitude of $\vec{d}$

$$WD = (F \cos \theta) \cdot d$$

$$WD = (\text{component of force along displacement}) \times (\text{displacement (magnitude)})$$

$$WD = F (d \cos \theta)$$

$$WD = \text{Force} \times \text{Component of displacement (magnitude) walong force}$$

1. If $\theta = 0^\circ \Rightarrow (WD)_F = Fd$
2. If $\theta = 90^\circ \Rightarrow (WD)_F = 0$
3. If $\theta = 180^\circ \Rightarrow (WD)_F = -Fd$
4. If $\theta < 90^\circ, (WD)_F > 0$
5. If $\theta > 90^\circ, (WD)_F < 0$

Q. If a force $\vec{F} = 3\hat{i} + 4\hat{j} + 6\hat{k}$ is acting on a particle such that particle moves from (1, 2, 3) to (7, 8, 5). Find WD by force $\vec{F}$.

$$\text{Sol. } \vec{F} = 3\hat{i} + 4\hat{j} + 6\hat{k}$$

$$\text{Displacement} = \text{change in position} = 6\hat{i} + 6\hat{j} + 2\hat{k}$$

$$(WD)_F = \vec{F} \cdot \vec{d}$$

$$= 18 + 24 + 12 = 54$$

Q. A constant force $\vec{F}$ of magnitude 100 N is applied at an angle 37° with horizontal on a block of mass 10 kg placed on ground. Find work done by force F and gravity when block move by 10 m.

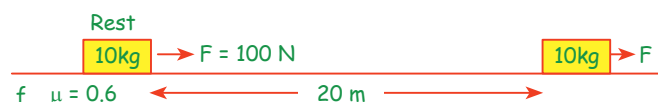
Sol.



$$\vec{F} = 80\hat{i} + 60\hat{j} \quad \vec{d} = 10\hat{i}$$

$$(WD) = \vec{F} \cdot \vec{d} = 800 + 0 = 800 \text{ J}$$

Q. Find the work done by forces F , f , mg and N if block move 20 m along + x axis



1 (WD) by $F = 100 \times 20 = 2000 \text{ J}$

2 (WD) gravity_(mg) = $0 \cdot (\because \vec{F} \perp \vec{d})$

3 (WD) by normal_(N) = 0

4 (WD) by normal_(f) = $-f_k d = -60 \times 20 = -1200$

$$\begin{aligned} (WD) \text{ by all force} &= W_g + W_N + W_f + W_F \\ &= 0 + 0 - 1200 + 2000 \\ &= +800 \text{ J} \end{aligned}$$

$$(WD) \text{ by net force} = + (F - f) \times d$$

↓
const

$$= (100 - 60) \times 20 = +800 \text{ J}$$

SKC

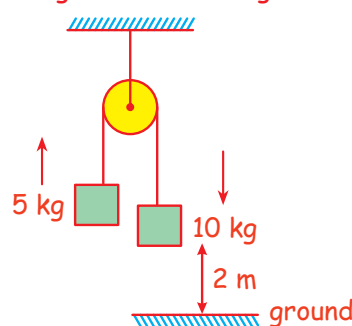


Yahan humne dekha (WD) by all the force nikalo ya (WD) by net force nikalo dono same hai.

Dekh bhai jab kisi force ka work done likh rahe hai to NSP nahi karni hai, NSP bole to Nayan sukh prapti matlab taada taadi nahi karni hai are mera matlab hai jiska force ka WD nikal rahe ho usi force par focus karo, dusri ki taraf dekhna bhi nahi hai.



Q. In given fig. system is released from rest. When 10 kg block reaches ground then find



- (i) Work done by gravity on 10 kg.
- (ii) Work done by gravity on 5 kg.
- (iii) Work done by tension on 10 kg.
- (iv) Work done by tension on 5 kg.

Sol. (i) $(W_g)_{10\text{kg}} = 10g \times 2 \cos 0^\circ = 200 \text{ J}$

(ii) $(W_g)_{5\text{kg}} = 5g \times 2 \times \cos 180^\circ = -100 \text{ J}$

(iii) $T = \frac{2m_1m_2}{m_1 + m_2}g = \frac{200}{3} \text{ N}$

$$(W_T)_{10\text{kg}} = \frac{200}{3} \times 2 \times \cos 180^\circ = \frac{-400}{3} \text{ J}$$

(iv) $(W_T)_{5\text{kg}} = \frac{200}{3} \times 2 \times \cos 0^\circ = \frac{400}{3} \text{ J}$

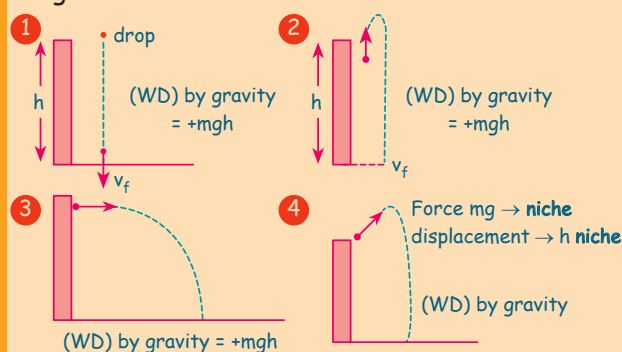
Net work done by tension is zero. Work done by internal tension i.e. (tension acting within system) on the system is always zero if the length of string remains constant.

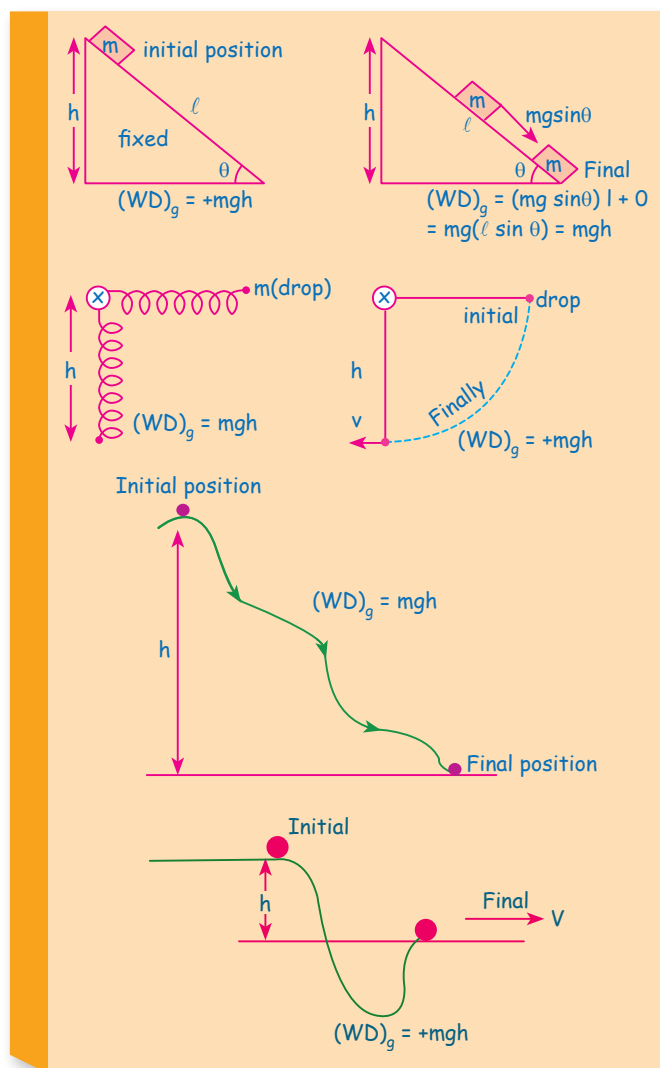


Kaam ka Dabba

Agar koi particle kaise bhi neche 'h' aya to work done by gravity = $+mgh$ hoga

Agar uper 'h' gaya hai to WD by gravity = $-mgh$ hoga





WORK DONE BY VARIABLE FORCE

Yahan aapko do prakar ke ques dekhne ko milenge ek Area calculation and doosra integration based since,

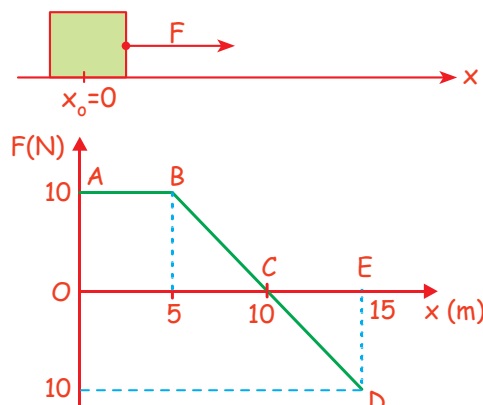
$WD = \int \vec{F} \cdot d\vec{s}$ so, F vs displacement (s) ke graph ka area under curve, WD by force F dega.



$WD = \int \vec{F} \cdot d\vec{s}$

Area = $\int F ds$

Q. A horizontal force F is used to pull a box placed on floor. Variation in the force with position coordinate x measured along the floor as shown in the graph.



(a) Calculate work done by the force in moving the box from $x = 0$ m to $x = 10$ m.

(b) Calculate work done by the force in moving the box from $x = 0$ m to $x = 15$ m.

Sol. In rectilinear motion work done by a force equals to area under the force-position graph.

(a) $W_{0 \rightarrow 10} = \text{Area of trapezium OABC} = 75 \text{ J}$

(b) $W_{0 \rightarrow 15} = \text{Area of trapezium OADE} - \text{Area of triangle CDE} = 50 \text{ J}$

Ab aapko only mathematics dekhne ko milegi so, maths ka dimag on kar do aur Sachin sir ka naam lekar nagin ko yaad karo bole to..... EX ko yaad karne ko nahi bol raha hoon i mean integration.



$$\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$$

$$(WD) \text{ by } F = \int \vec{F} \cdot d\vec{s} = \int_{x_i}^{x_f} F_x dx + \int_{y_i}^{y_f} F_y dy + \int_{z_i}^{z_f} F_z dz$$

Q. If particle move from $x = 0$ to $x = 2$. Find (WD) by $F = 3x^2$.

Sol. $\int \vec{F} \cdot d\vec{x} = \int_0^2 3x^2 dx = 3 \left[\frac{x^3}{3} \right]_0^2 = 2^3 - 0^3 = 8 \text{ J}$

Q. $\vec{F} = 2x\hat{i} + 2y\hat{j} + 4z^3\hat{k}$

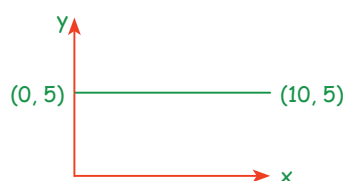
If particle move from (0, 0, 0) to (1, 2, 2), WD=?

Sol. $(WD)_F = \int_0^1 2x dx + \int_0^2 2y dy + \int_0^2 4z^3 dz$

$$= \left[\frac{2x^2}{2} \right]_0^1 + \left[\frac{2y^2}{2} \right]_0^2 + \left[\frac{4z^4}{4} \right]_0^2 = 1 + 4 + 16 = 21 \text{ J}$$

Q. $\vec{F} = 2x\hat{i} + 3x^2y\hat{j} = F_1\hat{i} + F_2\hat{j}$

If a particle moves parallel to x axis from (0, 5) to (10, 5) find WD by $\vec{F}$.



F_2 ka WD zero hai kyunki F_2 along y-axis hai aur displacement along x-axis



Sol. $(WD)_{F_1} = \int_0^{10} 2x dx = 100$

$(WD)_{F_2} = 0$

(No displacement in y- direction)

Q. If a particle moves from (0, 0) to (3, 4) on a straight along line $y = \frac{4x}{3}$ then find the WD by force by force

$\vec{F} = 2x\hat{i} + x^2y\hat{j}$

Sol. $(WD) = \int_0^3 2x dx + \int_0^4 x^2 y dy$

Ab ise directly integrate nahi kar sakte aur na hi x^2 ko bahar nikal sakte hai so, ab kya kare. Are kuch nahi karna x ki value y ke term me put kar do.



Arey pehle batana chahiye tha na

$= \left(2 \times \frac{9}{2} \right) + \int_0^4 \frac{9}{16} y^3 dy \quad \left[x = \frac{3}{4} y \right]$

$= 9 + \frac{9}{16} \frac{4^4}{4} = 9 + (9 \times 4) = 45$

Q. Find (WD) by force $F = 4x\hat{i} + y^2x\hat{j}$ if a particle move from (0, 0) to (2, 5) on the curve $y^2 = x$.

Sol. $WD = \int_0^2 4x dx + \int_0^5 y^2 y^2 dy = 633$

Q. Find the work done by force $\vec{F} = (xy\hat{i} + y\hat{j})N$ on particle, when it moves from origin to point (1, 1) along the path $y = x^2$, where x and y are in meters.

Sol. $W = \int F_x dx + \int F_y dy$

$W = \int xy dx + \int y dy$

$\Rightarrow y = x^2 \Rightarrow \frac{dy}{dx} = 2x$ or $dy = 2x dx$

$\Rightarrow W = \int_0^1 x^3 dx + \int_0^1 x^2 2x dx$

$= \frac{x^4}{4} \Big|_0^1 + \frac{2x^4}{4} \Big|_0^1 = \frac{1}{4} + \frac{2}{4} = \frac{3}{4} J$

$WD = \int \vec{F} \cdot d\vec{s}$

We know: $\vec{v} = \frac{d\vec{s}}{dt}$

$d\vec{s} = \vec{v} dt$

If $\vec{F}$ is $\perp^{\text{ar}} \vec{v}$

$WD = \int \vec{F} \cdot \vec{v} dt$

$\Rightarrow (WD)_F = 0$

Q. A force $\vec{F} = (3t\hat{i} + 5\hat{j})N$ acts on a body due to which its position varies as $\vec{S} = (2t^2\hat{i} - 5\hat{j})$. Work done by this force in first two seconds is:

Sol. $\frac{d\vec{S}}{dt} = 4t\hat{i}$

$d\vec{S} = 4t dt \hat{i}$

$W = \int \vec{F} \cdot d\vec{s}$

$= \int (3t\hat{i} + 5\hat{j}) \cdot (4t dt \hat{i})$

$= \int_0^2 12t^2 dt = \frac{12 \left[t^3 \right]_0^2}{3} = 32 J$

Work Done by Spring Force

$$\therefore \vec{F}_{sp} = -k\vec{x}$$

$$(WD)_{\text{spring force}} = \int \vec{F} d\vec{x} = \int_{x_i}^{x_f} -kx dx$$

$$(WD)_{\text{spring force}} = -\frac{1}{2}k(x_f^2 - x_i^2) = \frac{1}{2}k(x_i^2 - x_f^2)$$

Where $x_i \rightarrow$ Initial compression/elongation in spring from natural length

Where $x_f \rightarrow$ Final compression/elongation in spring from natural length

Ye ek bahut hi important result hai jo aapko yaad rakhna hai ki.

$$(WD)_{\text{spring force}} = -\frac{1}{2}k(x_f^2 - x_i^2)$$

Iski calculation me spring compress hai ya elongated hai koi fark nahi padta.



Q. If initial compression in spring is 2m. Find (WD) by spring force. If final compression in spring is 5m ($k = 1000 \text{ N/m}$).

$$\begin{aligned} \text{Sol. } (WD)_{sp} &= -1/2k(x_f^2 - x_i^2) \\ &= -1/2 \times 1000 \times (5^2 - 2^2) \\ &= -500 \times 21 \\ &= -10500 \text{ J} \end{aligned}$$

Q. Find (WD) by spring force if spring is compressed by x_0 from natural length.

$$\begin{aligned} \text{Sol. } (WD)_{sp} &= -1/2k(x_0^2 - 0^2) \\ &= -1/2(kx_0^2) \end{aligned}$$

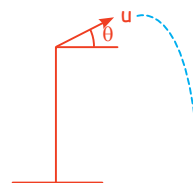
WORK ENERGY THEOREM (WET)

WD by all force = change in K.E = $\Delta K.E = (K.E)_{\text{final}} - (K.E)_{\text{Initial}}$

Yah bahut hi zyada important theorem hai jo ek karod bar use hogi.



Q. Find final speed of particle just before it hits the ground

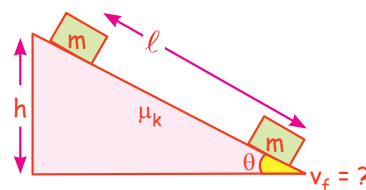


$$\text{Sol. } (WD)_g = \Delta K.E = K_f - K_i$$

$$mgh = \frac{1}{2}mv_f^2 - \frac{1}{2}mu^2$$

$$v_f^2 = u^2 + 2gh$$

Q. Find final velocity V_f if block starts from rest

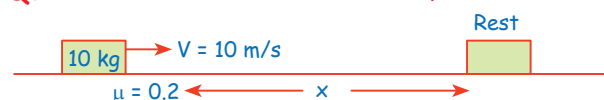


$$\text{Sol. } W_g + W_N + W_f = \Delta K.E$$

$$+mgh + 0 - (\mu mg \cos \theta \ell) = \frac{1}{2}mv_f^2 - 0$$

Solve for V_f .

Q. Calculate distance covered before block stops.



Sol. M-1

$$W_g + W_N + W_f = \Delta K.E$$

$$0 + 0 - \mu mgx = 0 - \frac{1}{2}mV^2$$

$$0.2 \times 10 \times x = \frac{1}{2} \times 10 \times 10$$

$$x = \frac{50}{2} \therefore x = 25 \text{ m}$$

M-2

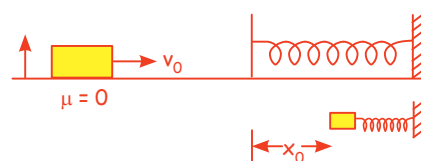
$$v^2 = u^2 + 2as$$

$$0 = 10^2 - 2 \times \mu gx$$

$$2\mu gx = 100$$

$$x = \frac{100}{2 \times 0.2 \times 10} = 25$$

Q. Find maximum compression in spring



Sol. $W_g + W_N + W_{sp} = \Delta KE$

$$0 + 0 + -1/2 k (x_0^2 - 0^2) = 0 - 1/2 m v_0^2$$

$$x_0 = v_0 \sqrt{\frac{m}{k}}$$

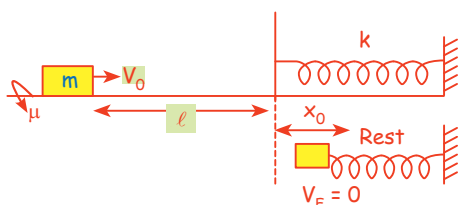
Maximum compression in spring tab hoga jab block rest par aa jayega. Matlab $v_f = 0$ aur yah concept karodo bar use hoga, 10-12 question to mai hi add kar de raha hu.



Maan lo in above ques friction bhi diya hota..... to kar loge na..... only friction ka WD hi to likhna hai.



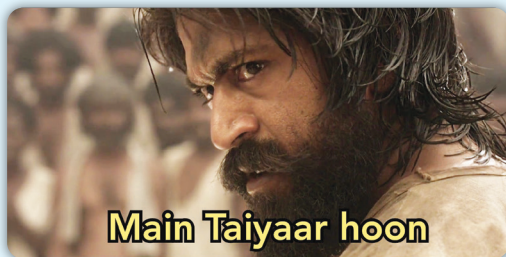
Q. Surface is rough. Find max compression x_0



Sol. $W_g + W_N + W_f + W_{sp} = \Delta KE$

$$0 + 0 - \mu mg(l + x_0) - 1/2 k (x_0^2 - 0^2) = 0 - 1/2 m v_0^2$$

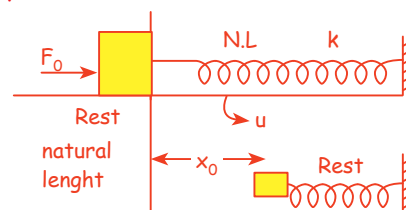
Ab dhyan se suno.... we have limited no. of pages but hum question practice me koi kami nahi chhodenge.... **ye lo pen** aur jitne question mai attach kar raha hu that's sufficient to build your concept so rough copy par khud se inhen zarur solve kare are you ready....



Main Taiyaar hoon

Mujhe insta (Saleem.nitt) par update kare only if you have a account.

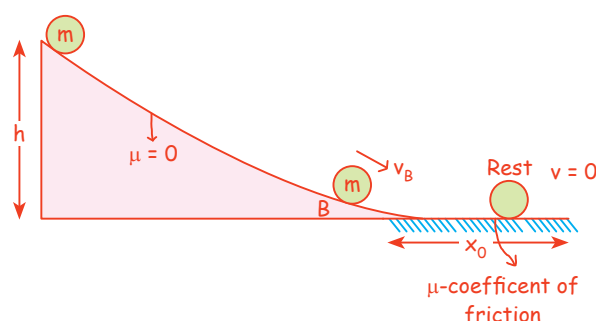
Q. Contant force F_0 acts on block. Find maximum compression.



Sol. $W_g + W_N + W_f + W_{sp} + W_F = \Delta KE$

$$0 + 0 - \mu mg x_0 - 1/2 k (x_0^2 - 0^2) + F_0 x_0 = 0 - 0$$

Q. Horizontal surface is rough. Find velocity at B & x_0 .



For velocity at B

$$W_g + W_N = \Delta KE$$

$$mgh + 0 = 1/2 m v_B^2 - 0$$

$$v_B = \sqrt{2gh}$$

To find x_0

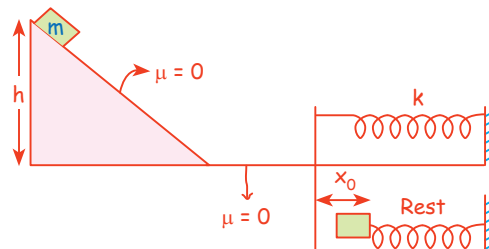
$$W_g + W_N + W_f = \Delta KE$$

$$mgh + (0 - \mu mg x_0) = 0 - 0$$

$$mgh = \mu mg x_0$$

$$x_0 = h/\mu$$

Q. Find max compression in spring

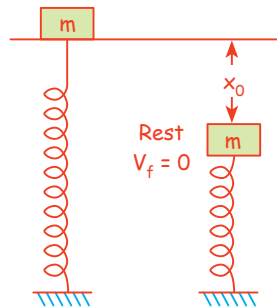


Sol. $W_g + W_N + W_f + W_{sp} = \Delta KE$

$$mgh + 0 + 0 - 1/2 k (x_0^2 - 0^2) = 0 - 0$$

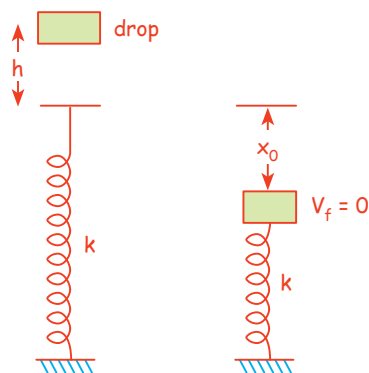
$$x_0 = \sqrt{\frac{2mgh}{k}}$$

- Q. Find maximum compression in spring.
Spring is initially in its natural length.



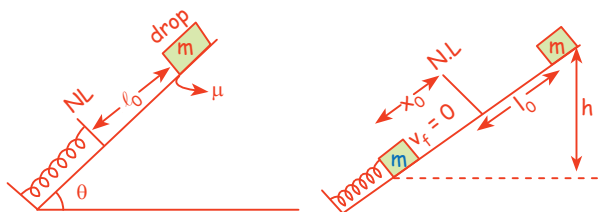
Sol. $W_g + W_{SP} = \Delta KE$
 $mgx_0 - 1/2k(x_0^2 - 0) = 0 - 0$
 $x = \frac{2mg}{k}$

- Q. Find maximum compression. At max elongation $v = 0$



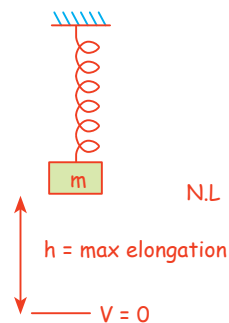
Sol. $W_g + W_{SP} = \Delta KE$
 $mg(h + x_0) - 1/2k(x_0^2 - 0^2) = 0 - 0$
 $\Rightarrow$ Solve for x_0 .

- Q. Find maximum compression in spring.

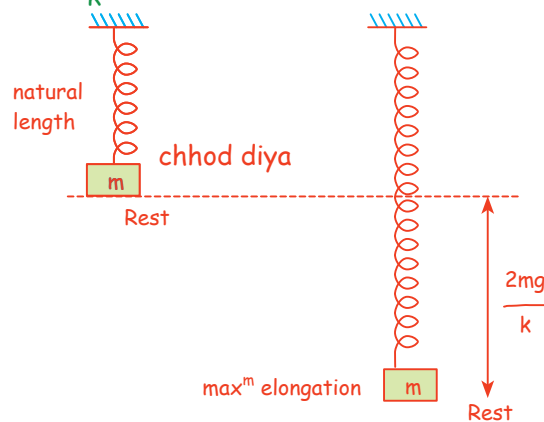


Sol. $W_g + W_N + W_{SP} + W_f = \Delta KE$
 $mg(l_0 + x_0) \sin \theta + 0 - \frac{1}{2}k(x_0^2 - 0)$
 $- \mu mg \cos \theta (l_0 + x_0)$
 $= 0 - 0$

- Q. Find maximum elongation. When a block of mass m which is attached to a spring, is released from its Natural length.



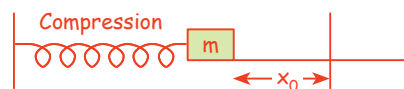
Sol. $W_g + W_{SP} + W_N = \Delta KE$
 $mgh - 1/2k(h^2 - 0^2) + 0 = 0 - 0$
 $mgh = 1/2kh^2$
 $h = \frac{2mg}{k}$



Yaad rakhna yahan natural length se equilibrium position, mg/k doori par hai jahan velocity max. hogi.

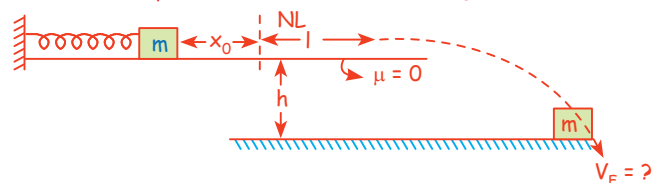


- Q. Find K.E when block is released from rest.



Sol. $W_g + W_N + W_F + W_{SP} = \Delta KE$
 $0 + 0 + 0 - 1/2k(0^2 - x_0^2) = 1/2mV^2$
 $V = \sqrt{\frac{kx_0^2}{m}}$

- Q. Find V_f , initial compression is x_0 .

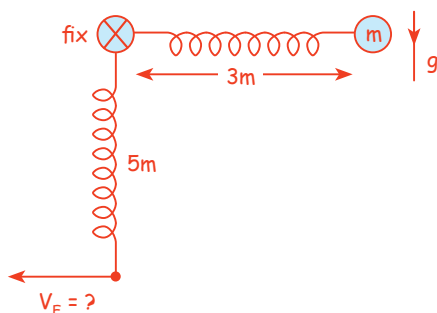


Sol. $W_g + W_N + W_f + W_{SP} = \Delta KE$
 $+ mgh + 0 + 0 - 1/2 k (0^2 - x_0^2) = 1/2 m V_F^2 - 0$

If friction (μ) is present

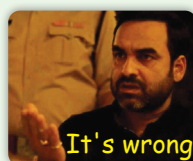
$W_g + W_N + W_f + W_{SP} = \Delta KE$
 $+ mgh + 0 - \mu mg (x_0 + l) - 1/2 K (0^2 - x_0^2)$
 $= 1/2 m V_f^2 - 0$

Q. A particle is released from rest in following diagram find velocity V_F if natural length of the spring is 2m.



Sol.

$(mg)5 - 1/2 K [5^2 - 3^2] = 1/2 m V_f^2$ (wrong)



$mg5 - 1/2 k [5^2 - 1^2] = 1/2 m V_f^2$

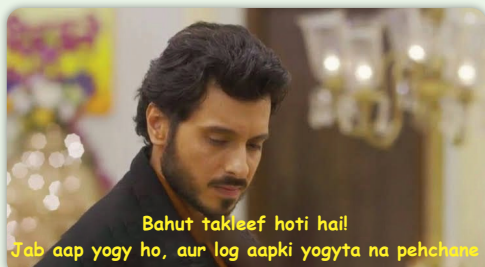
(It's also wrong)



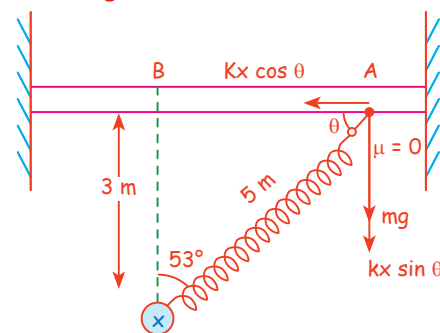
x_f and x_i are elongation from natural length

$mg5 - 1/2 k [3^2 - 1^2] = 1/2 m V_f^2$

(Now it's correct)



Q. Find velocity of ring when it reaches at B, natural length = 2m



Sol. $x_i = 5 - 2 = 3 \text{ m}$

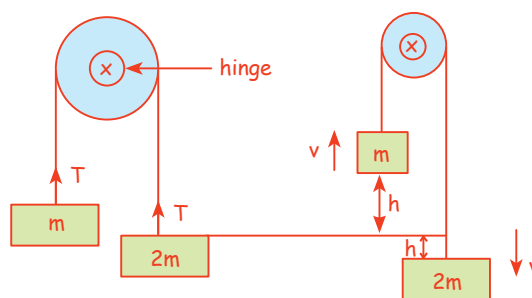
$x_f = 3 - 2 = 1 \text{ m}$

$A \rightarrow B$

$W_g + W_N + W_{sp} + W_f = \Delta KE$

$0 + 0 - 1/2 k (1^2 - 3^2) + 0 = 1/2 m V_B^2 - 0$

Q. Blocks are released from rest, find speed of each block when 2m block goes down by h.



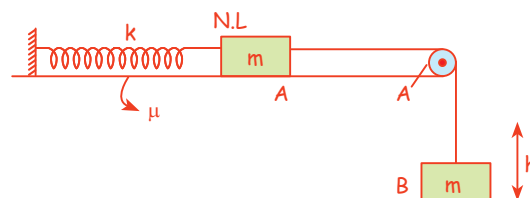
Sol. $W_g + W_T + W_{hinge} = \Delta KE$

$(+2mgh - mgh) + (-Th + Th) + 0$

$= \left(\frac{1}{2} 2mv^2 + \frac{1}{2} mv^2 \right) - 0$

$mgh = \frac{3mv^2}{2} \quad v = \sqrt{\frac{2gh}{3}}$

Q. Find speed of both block if B goes down by 'h'

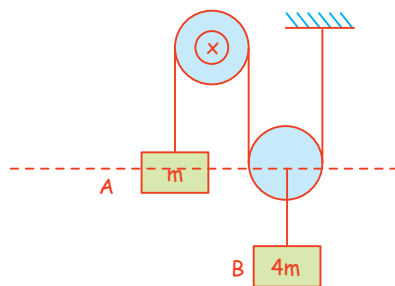


Sol. $W_{SP} + W_f + W_T + W_g + W_{hinge} + W_N = \Delta KE$

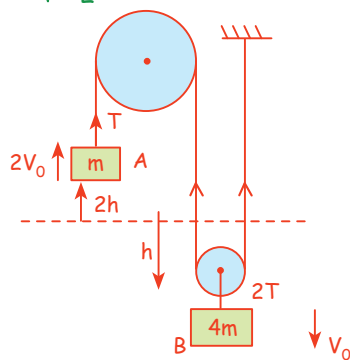
$- \frac{1}{2} k (h^2 - 0^2) - \mu mgh + 0 + (mgh + 0) + 0$

$= \left(\frac{1}{2} mv^2 + \frac{1}{2} mv^2 \right) - 0$

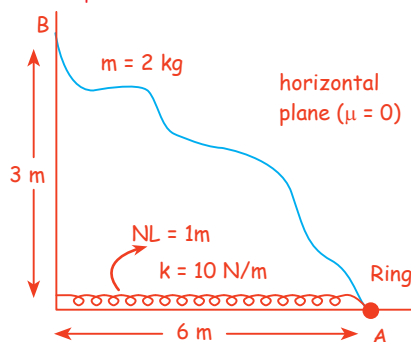
- Q. Find speed of both blocks when block B reaches distance h in downward direction



Sol. $w_g + w_T + w_{\text{hinge}} = \Delta KE$
 $(+4mgh - mg2h) = \frac{1}{2} 4mV_0^2 + \frac{1}{2} m^2 V_0^2 - 0$
 $2mgh = 4mv_0^2 \Rightarrow V_0 = \sqrt{\frac{gh}{2}}$
 $V_A = 2V_0 = \sqrt{2gh}$
 $V_B = V_0 = \sqrt{\frac{gh}{2}}$

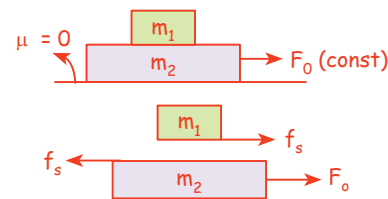


- Q. When an unknown variable force 'F' is applied on ring lying on horizontal plane such that when ring reaches at point B it acquires speed 5 m/s. Natural length of spring is 1 m and $k = 10 \text{ N/m}$. Find $(WD)_F = ?$



Sol. $W_g + W_N + W_{sp} + W_F = \Delta KE$
 $0 + 0 - \frac{1}{2} \times 10 (2^2 - 5^2) + w_F = \left(\frac{1}{2} \times 2 \times 5^2 - 0 \right)$
 $\frac{1}{2} \times 10 \times 21 + w_F = \frac{50}{2}$
 $w_F = -80 \text{ J}$

- Q. In the following problem if both blocks move together without slipping with same acceleration by distance x . Find net WD by friction force.



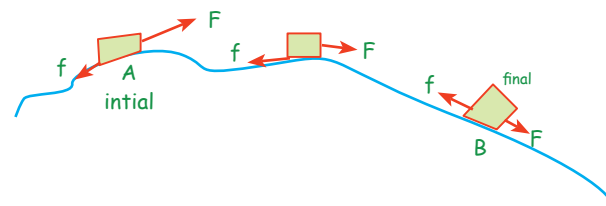
Sol. (WD) by friction on $m_1 = + f_s \cdot x$
(WD) by friction of $m_2 = - f_s x$
Net (WD) by friction $= f_s x + (-f_s \cdot x) = 0$
Work done by action reaction pair of constraint forces such as static friction, normal reaction and tension in the string is always zero when they are internal forces.

SKC

- WD by static friction can be positive / negative or zero but net WD by static friction always zero.
- WD by kinetic friction can be positive / negative or zero but net WD by static friction always negative.



WD BY FORCE IF F IS ALWAYS PARALLEL TO DISPLACEMENT OR F IS ALWAYS TANGENT TO THE PATH

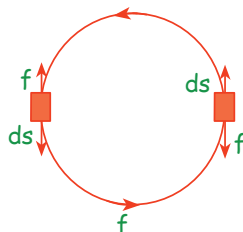


- Q. Find (WD) by force F from A to B if magnitude of force is constant and direction is parallel to displacement at each point on path.

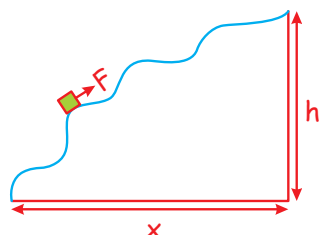
Sol. $F \rightarrow$ always tangent/parallel has Displacement
 $\theta = 0 \Rightarrow (WD)_F = F \times \text{distance} = F \times (\text{path length})$
 $F \rightarrow$ always Anti parallel, $\theta = 180^\circ$, displacement
 $(WD)_F = -F \times (\text{Distance})$

Q. A block move on horizontal rough surface in a circular path of radius R as shown in diagram Find (WD) by friction in one complete revolution.

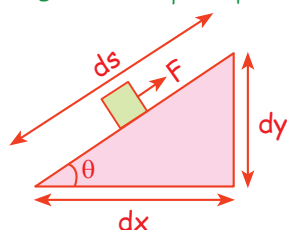
Sol. Force always antiparallel to displacement ($\theta = 180^\circ$)
 $w = -f \cdot (2\pi R)$ path length
 $w = -\mu mg \cdot 2\pi R$



Q. A body of mass m is slowly hauled up the rough hill by a force F which at each point is directed along a tangent to the hill. Work done by the force.



Sol. $w_g + w_N + w_f + w_F = \Delta KE$
 $-mgh + 0 + W_f + W_F = 0$



$$ds \cos \theta = dx$$

$$\int ds \cos \theta = \int dx = x$$

and $|W_f| = \int (\mu mg \cos \theta) ds$
 $= \mu mg \int ds \cos \theta = \mu mg \int dx = \mu mgx$
 $-mgh + 0 - \mu mgx + W_F = 0$
 $W_F = mgh + \mu mgx$

Ab kuchh aise sawal solve karte hai jinme maths aur integration involve hoti hai ye question jyadatar Mains mein puche jate hai.



Q. If a particle of mass 2kg moving on x-axis such that $x = t^4$ under the action of several force. Find WD by all the force from $t = 0$ to $t = 2$ sec

Sol. **m-1** $x = t^4$,
 $v = 4t^3$
 $a = 12t^2$
 $F = ma = 2 \times 12t^2$
 $F = 24t^2$



$$WD = \int \vec{F} \cdot \vec{v} dt$$

$$= \int 24t^2 \cdot 4t^3 dt$$

$$= 24 \times 4 \times \frac{2^6}{6} = 1024$$

Ye accha method nahi hai sidha WET lagao.



m-2

$$x = t^4$$

$$v = 4t^3$$

$$t = 0, v_i = 0$$

$$t = 2 = 4 \times 2^3 = 32 = v_f$$

$$(WD) \text{ all force} = K_f - K_i$$

$$= 1/2 \times 2 \times 32^2 - 0 = 1024.$$



Q. Find (WD) by all forces from $t = 0$ to $t = 3$ if $m = 2\text{kg}$, $v = 2t^2$.

Sol. $t = 0$ $v = 0$, $(KE)_i = 0$
 $t = 2$ $v = 2 \times 3^2 = 2 \times 9 = 18$
 $(KE)_f = 1/2 \times 2 \times 18^2 = 324$
 $(WD) \text{ all force} = 324 - 0$

Q. A particle (2kg) start from rest from origin & start moving on x-axis its speed v/s x relation is find (WD) by all force from $x = 0$ to $v = 4x^{3/2}$, $x = 4$.

Sol. $x = 0$, $v = 0 \Rightarrow (KE)_i = 0$
 $x = 4$, $v = 4 \times 4^{3/2} = 4 \times 8 = 32$ $(KE)_f = 0$
 $(WD) \text{ all force} = 1/2 \times 2 \times 32^2 - 0$ $(KE)_f = (32)^2$

Q. A particle (1 kg) move on x-axis under the action of several force such that $x = \sqrt{t} - 2$.

Find (WD) by all force from $t = 1 \rightarrow t = 2$ sec.

Sol. $v = \frac{dx}{dt} = \frac{1}{2\sqrt{t}}$

$$t = 1, v = 1/2 \quad KE_i = 1/2 \times 1 \times 1/4$$

$$t = 2, v = \frac{1}{2\sqrt{2}} \quad KE_f = \frac{1}{2} \times 1 \times 1/8$$

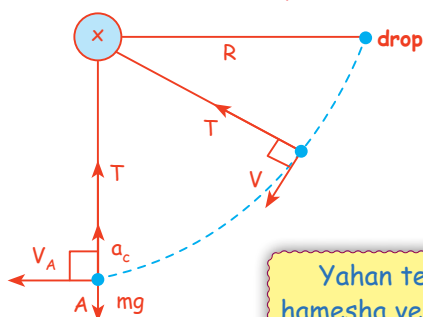
$$W = \Delta KE = \frac{1}{16} - \frac{1}{8} = -\frac{1}{16} \text{ J}$$

Ab hum WET + circular motion ke mix question ki practice karenge. In question mein bas ye yaad rakhna humein mostly do eqn ke saath khelna hai.

1. Work energy theorem ki eqn
2. Circular motion ki centripetal force ki eqn



Q. The particle is released from rest as shown in figure. Find velocity of particle, tension, centripetal and tangential accelerations. (a_t , a_c) when it reaches at lowest point.



Yahan tension hamesha velocity ke perpendicular hai so, WD by tension = 0 (Yahan acche se samajh lo bahut bar use hoga)

Sol. $w_g + w_T = \Delta KE$

$$+mgR + 0 = \frac{1}{2} mV_A^2 - 0$$

$$V_A = \sqrt{2gR}$$

Tension when reaches at lowest pt

$$T_A - mg = ma_c$$

$$T_A - mg = \frac{mV^2}{R}$$

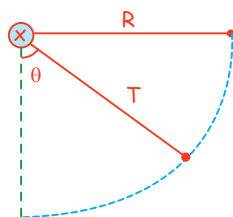
$$T_A - mg = \frac{m2gR}{R}$$

$$T_A = 3mg$$

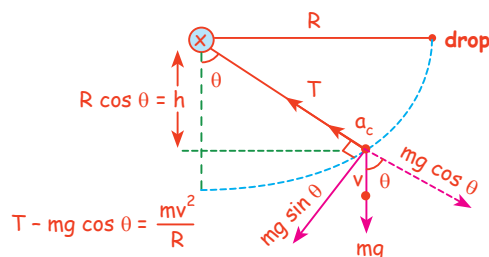
$$a_c = \frac{v^2}{R} = \frac{2gR}{R} = 2g$$

$$a_t = 0$$

Q. Find speed of ball and Tension in string T when string makes angle θ with vertical. Take $\theta = 60^\circ$, $m = 2\text{kg}$, $r = 10\text{m}$.



Sol.



$$w_g + w_T = \Delta KE$$

$$mgh = \frac{1}{2} mv^2 - 0$$

$$V^2 = 2gh$$

$$V = \sqrt{2gR \cos \theta} = \sqrt{2 \times 10 \times 10 \cos 60^\circ} = 10 \text{ m/s}$$

$$mg \sin \theta = ma_t$$

$$a_t = g \sin \theta$$

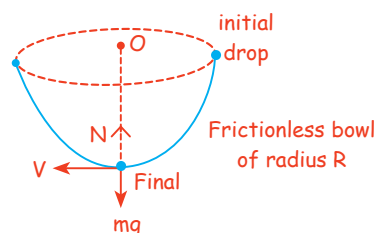
Now let's write circular motion eqn

$$T - mg \cos 60^\circ = \frac{mV^2}{R}$$

$$T - 2 \times 10 \times \frac{1}{2} = \frac{2 \times 100}{10}$$

$$\therefore T = 30 \text{ N}$$

Q. A particle is dropped from a point inside a smooth fixed hemisphere as shown in fig. When particle reaches at lowest point find Speed (V) and Normal (N).



Sol. $W_f + W_g + W_N = \Delta KE$

$$0 + mgR + 0 = \frac{1}{2} mV^2 - 0$$

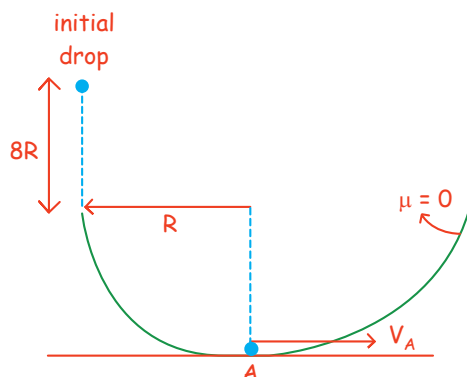
$$V = \sqrt{2gR}$$

$$N - mg = \frac{mV^2}{R} = N = 3mg$$

Yaar last question ki tarah yahan agar kisi angle par v , N pucha jaye to nikal loge na..... same as it is eqn banegi bas tension ki jagah normal aa jayega aur WD by normal is 0.



Q. Find speed V_A of particle when it reaches at lowest point A.

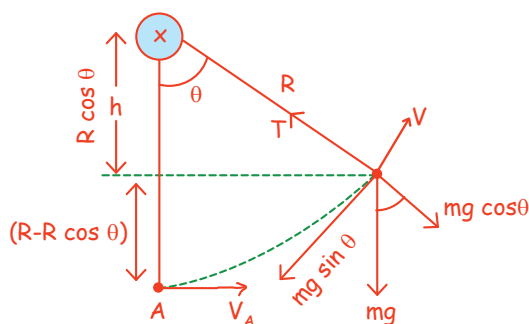


Sol. $W_g + W_N = \Delta KE$

$$mg9R + 0 = \frac{1}{2}mv^2 - 0$$

$$V_A = \sqrt{18gh}$$

Q. Bob of pendulum is given velocity V_A at lowest position. Find Tension when string makes angle θ (if $\theta < 90$). Find min velocity of particle to perform vertical circular motion at C.



Sol. $w_g + w_T = \Delta KE$ (WET)

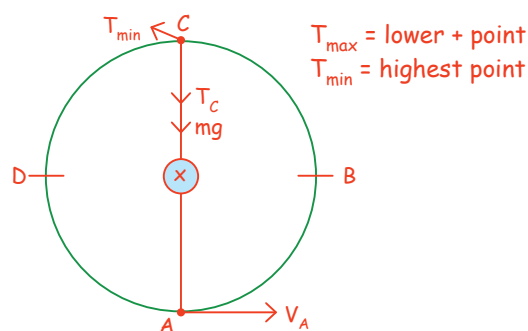
$$-mg(R - R \cos \theta) + 0 = \frac{1}{2}mv^2 - \frac{1}{2}mv_A^2$$

$$T - mg \cos \theta = \frac{mV^2}{R} \text{ (Centripetal force eqn)}$$

or

$$T = mg \cos \theta + \frac{mV^2}{R}$$

Jaise jaise ball upar jayegi tension kam hoti jayegi, for minimum velocity at C, highest point par tension zero kar do. Aisa karne par mg ne ijjat bacha li bole to centripital force requirement fullfill kar di.



$$\text{at } C \Rightarrow mg + T_C = \frac{mV_C^2}{R}$$

$$T_C \rightarrow \min = 0$$

$$mg = \frac{mV_C^2}{R}$$

$$V_C = \sqrt{gR}$$

(C $\rightarrow$ A)

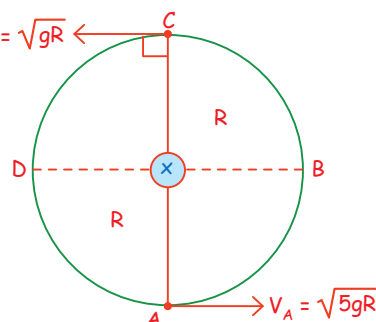
$$w_g + w_T = \Delta KE$$

$$mg2R + 0 = \frac{1}{2}mV_A^2 - \frac{1}{2}m(gR)$$

$$4mgR = mV_A^2 - mgR$$

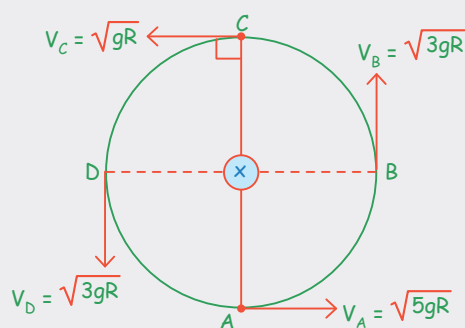
$$\text{then if } V_C = \sqrt{gR}, V_A = \sqrt{5gR}$$

$$\text{If } V_C = \sqrt{gR} \text{ then } V_A = \sqrt{5gR}$$



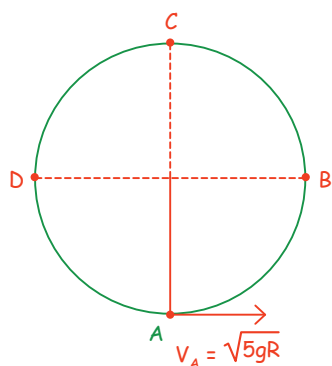
This is Very Important Part

Minimum velocity at A to complete vertical circular motion is $\sqrt{5gR}$, is case mein remember hamne at highest point $T_c = 0$ kiya, mg ne ijjat bachai, $V_b = \sqrt{3gR}$ aur $V_c = \sqrt{gR}$



Ab aapko har point par V, T, a_r , a_c nikalna aana chahiye main mast tareeke se har ek-ek cheez ko solve karke represent kar de raha hoon. plz inhe khud se zaroor solve karein cost same ko lekar bhai publisher se mein lad loonga.

Q. Find T, v, a_t, a_c, a_{net} at A, B, C, D points in given case.



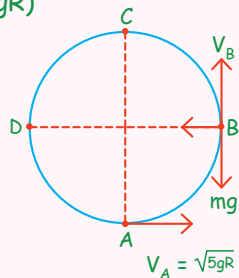
Sol.

At point 'A'

$$\begin{aligned} \star T_A - mg &= \frac{mV_A^2}{R} \\ T_A - mg &= \frac{m5gR}{R} \\ T_A &= 6mg \\ \star a_c &= \frac{V^2}{R} = \frac{5gR}{R} = 5g \\ \star a_t &= 0 \\ \star a_{net} &= \sqrt{a_t^2 + a_c^2} = 5g \\ \star V_A &= \sqrt{5gR} \end{aligned}$$

At point 'B' and 'D'

$$\begin{aligned} \star V_B : \text{WET from } A \rightarrow B \\ w_g + w_T &= \Delta KE \\ -mgR + 0 &= \frac{1}{2} mV_B^2 - \frac{1}{2} m(5gR) \\ V_B &= \sqrt{3gR} \\ \star a_c : \frac{V^2}{R} &= \frac{\sqrt{(3gR)^2}}{R} = 3g \\ \star T_B &= \frac{mV^2}{R} = \frac{m3gR}{R} = 3mg \\ \star mg &= ma_t \\ a_t &= g \text{ (downward)} \\ \star a_{net} &= \sqrt{(3g)^2 + (g)^2} = g\sqrt{10} \end{aligned}$$

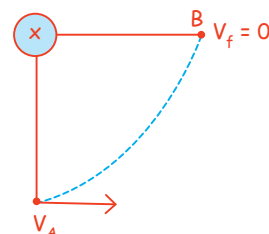


At point 'C'

$$\begin{aligned} \star V_C : \text{apply WET from } A \rightarrow C \\ w_g + w_T &= \Delta KE \\ -mg2R + 0 &= \frac{1}{2} mV_C^2 - \frac{1}{2} m5gR; V_C = \sqrt{gR} \\ \star T_C : T + mg &= \frac{mV^2}{R} \\ T + mg &= \frac{mgR}{R} \Rightarrow T_C = 0 \\ \star a_c &= \frac{V^2}{R} = \frac{gR}{R} = g \\ \star a_t &= 0 \\ \star a_{net} &= \sqrt{(g)^2 + (0)^2} = g \end{aligned}$$

	A	B, D	C	P(general point)
Velocity	$\sqrt{5gl}$	$\sqrt{3gl}$	$\sqrt{gl}$	$\sqrt{gl(3+2\cos\theta)}$
Tension	$6mg$	$3mg$	0	$3mg(1+\cos\theta)$
Potential Energy	0	$mg l$	$2mg l$	$mg l(1-\cos\theta)$
Radial acceleration	$5g$	$3g$	g	$g(3+2\cos\theta)$
Tangential acceleration	0	g	0	$g \sin\theta$

Q. Find velocity at A (lowest point) so that, string become horizontal & particle comes to rest at B.



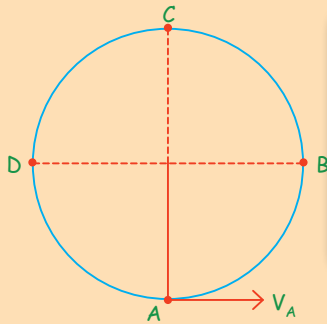
Sol. $w_g + w_T = \Delta KE$

$$-mgR + 0 = 0 - \frac{1}{2} mV_A^2, V_A < \sqrt{2gR}$$



Kaam ka Dabba

Particle given horizontal velocity at A

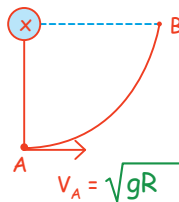


Ye result zaroor yaad kar lena bcz iske baad aapko pehle se pata hoga ki particle ka motion kya hoga.

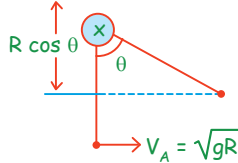


- ★ $V_A < \sqrt{2gR}$ = particle comes to rest before 'B' ($v = 0$ between A & B)
- ★ $V_A = \sqrt{2gR}$ = particle comes to rest at 'B'
- ★ $V_A = \sqrt{5gR}$ = vertical circular motion just completed ($T_C = 0$, $V_C = \sqrt{gR}$)
- ★ $V_A > \sqrt{5gR} \Rightarrow$ chamchamata vertical circular motion chamchamata $T_C \neq 0$
- ★ $\sqrt{2gR} < V_A < \sqrt{5gR} \Rightarrow$ somewhere between B & C, T will be zero and subsequently motion will be projectile.

- Q. 1. Find the 'maximum' height attained by particle.
2. Find maximum angle by string with vertical.



Sol. $w_g + w_T = \Delta KE$



$$-mg(R - R \cos \theta) + 0 = 0 - \frac{1}{2} m (\sqrt{gR})^2$$

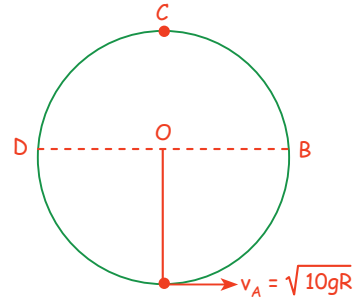
$$\cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

$$h = R - R \cos 60^\circ = R/2$$

Solve above question again if $V_A = \sqrt{1.5gR}$

$$\theta = \cos^{-1}(1/4) \text{ (verify this on your own)}$$

Q. Find T, v at A, B, C.



Sol.

★ at 'A'

$$T_A: T_A - mg = \frac{m10gR}{R}$$

$$T_A = 11mg$$

★ at 'B'

Applying WET from A $\rightarrow$ B

$$-mgR = \frac{1}{2}mv_B^2 - \frac{1}{2}m(\sqrt{10gR})^2$$

$$v_B = \sqrt{8gR}$$

$$T_B = \frac{mv_B^2}{R} \therefore \frac{m8gR}{R} = 8mg$$

★ at 'C'

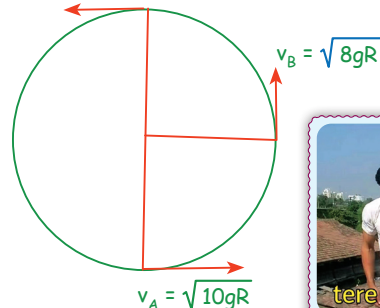
$$-mgh + 0 = \frac{1}{2}mv_C^2 - \frac{1}{2}m10gR$$

$$v_C = \sqrt{6gR}$$

$$T_C + mg = \frac{mv_C^2}{R}$$

$$T_C = 5mg$$

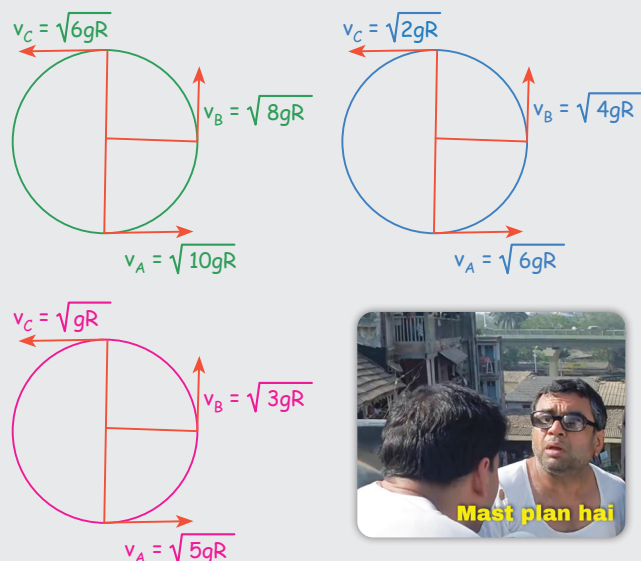
$$v_C = \sqrt{6gR}$$



Find $\frac{T_{\max}}{T_{\min}} = ? \Rightarrow \frac{T_{\max}}{T_{\min}} = \frac{T_A}{T_C}$



Dekh bhai ye pattern pehchanne ki koshish karo agar A par velocity $\sqrt{10gR}$ hai to B par velocity $\sqrt{8gR}$ aayi aur C par velocity $\sqrt{6gR}$ aayi matlab underroot ke andar ki digit do kam hoti ja rahi hai (minus two)... Abe samajh aaya le aur example dekh.



Q. A small bob tied at one end of a thin string of length 1m is describing a vertical circle so that the maximum and minimum tension in the string are in the ratio 5 : 1. The velocity of the bob at the highest position is _____ m/s. (Take $g = 10 \text{ m/s}^2$)

Sol. Let v is velocity at highest position.

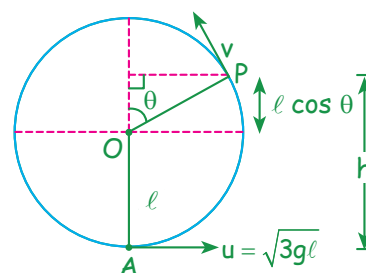
$$T_{\max} = 5T_{\min}$$

$$\Rightarrow \frac{mg + \frac{m(v^2 + 4gl)}{l}}{5 \frac{mv^2}{l} - mg} \Rightarrow 4 \cdot \frac{v^2}{l} = 10g \Rightarrow \frac{T_{\max}}{T_{\min}} = 5$$

$$\Rightarrow v = \sqrt{\frac{5}{2}gl} = \sqrt{\frac{5}{2} \times 10 \times 1} = 5 \text{ m/s}$$

Q. A bob of a stationary pendulum is given a sharp hit to impart it a horizontal speed of $\sqrt{3gl}$. Find the angle rotated by the string before it becomes slack.

Sol. Suppose the string becomes slack at point P. Let the bob rise to a height h .
 $h = l + l \cos \theta$



From the work-energy theorem,

$$\frac{1}{2}mv^2 - \frac{1}{2}mu^2 = -mgh$$

$$v^2 = u^2 - 2g(l + l \cos \theta) \quad \dots(i)$$

$$\text{Again, } \frac{mv^2}{l} = mg \cos \theta$$

$$v^2 = l g \cos \theta \quad \dots(ii)$$

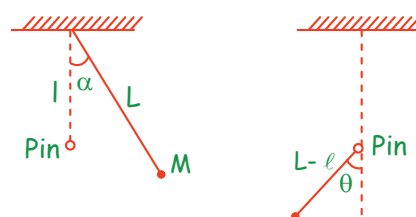
Using equation (i) and (ii) and the value of u , we get,

$$g l \cos \theta = 3g l - 2g l - 2g l \cos \theta$$

$$3 \cos \theta = 1$$

$$\cos \theta = \frac{1}{3}$$

Q. A simple pendulum consisting of a mass M attached to a string of length $L = 7\text{m}$ is released from rest at an angle $\alpha = 37^\circ$. A pin is located at a distance $\ell = 4\text{m}$ below the pivot point. When the pendulum swings down, the string hits the pin as shown in figure. The maximum angle which the string makes with the vertical after hitting the pin is θ . Find the value of $\cos \theta$.



Sol. Since the pendulum started with no kinetic energy, conservation of energy implies that the potential energy at $\theta_{\max}$ must be equal to the original potential energy, i.e., the vertical position will be same. Therefore,

$$mg \cos \alpha = mg [l + (L - l) \cos \theta]$$

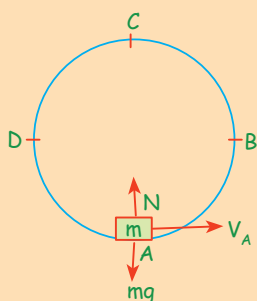
$$\Rightarrow \cos \theta = \frac{L \cos \alpha - l}{L - l}$$

$$\Rightarrow \theta = \cos^{-1} \left[\frac{L \cos \alpha - l}{L - l} \right]$$



Kaam ka Dabba

Particle given horizontal velocity at A

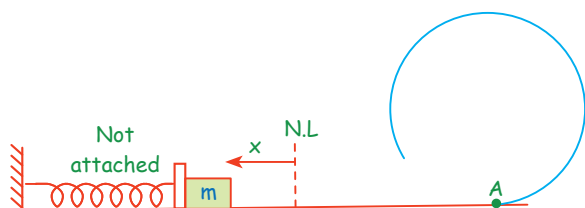


Ye result bilkul pehle jaise hai bas tension ki jagah ab normal aa gaya aur bhai ye VCM hai horizontal nahi be carefull.



- ★ $V_A < \sqrt{2gR}$ = particle comes to rest before 'B' ($v = 0$ between A & B)
- ★ $V_A = \sqrt{2gR}$ = particle comes to rest at 'B'
- ★ $V_A = \sqrt{5gR}$ = vertical circular motion just completed ($N_C = 0$, $V_C = \sqrt{gR}$)
- ★ $V_A > \sqrt{5gR} \Rightarrow$ chamchamata vertical circular motion
chamchamata $N_C \neq 0$
- ★ $\sqrt{2gR} < V_A < \sqrt{5gR} \Rightarrow$ somewhere between B & C, N will zero and subsequently motion will be projectile.

Q. Block is released from rest in following fig. Find minimum compression in spring so that block completes circle. Block not attached to spring.



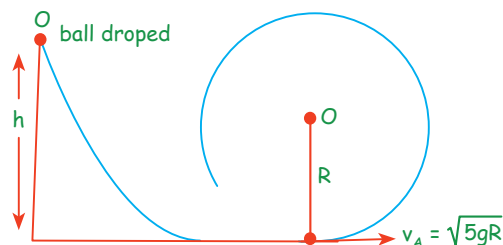
Sol. A par minimum velocity $\sqrt{5gR}$ honi chahiye

$$W_{SP} + W_g + W_N + W_f = \Delta KE$$

$$-\frac{1}{2}k(0 - x^2) + 0 + 0 + 0 = \frac{1}{2}m(\sqrt{5gR})^2 - 0$$

$$\frac{1}{2}kx^2 = \frac{1}{2}m5gR \quad x = \frac{\sqrt{5gRm}}{\sqrt{k}}$$

Q. Find minimum value of h to complete circle.

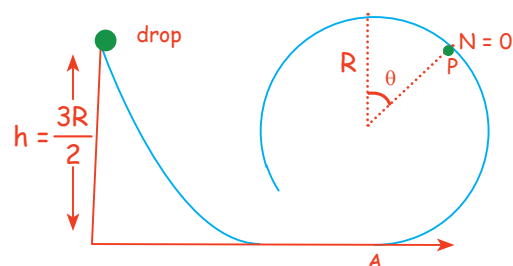


Sol. $W_g + W_N = \Delta KE$

$$mgh + 0 = \frac{1}{2}m(5gR) - 0$$

$$h = \frac{5R}{2} = h_{\min}$$

Q. If particle loose contact at point P find θ .

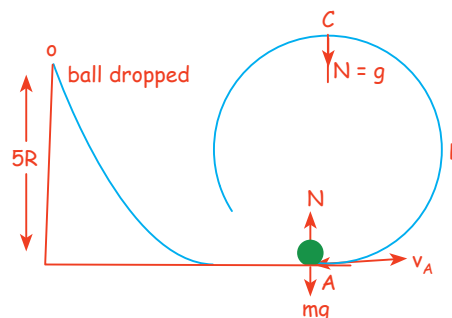


Sol. $mg\left(\frac{3R}{2}\right) + 0 = \frac{1}{2}mv_A^2 - 0$

$$v_A = \sqrt{3gR} \Rightarrow \cos \theta = \frac{1}{3}$$

H.W. If in above ques. $h = 7R/4$. Find θ (ans. 60°)

Q. Find force applied by track when particle reaches at A & C



Sol. $mg(5R) + 0 = \frac{1}{2}mv_A^2 - 0$

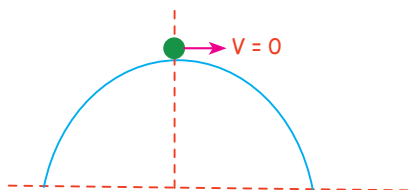
$$v_A = \sqrt{2g(5R)} = v_A = \sqrt{10gR}$$

$$N_A - mg = \frac{mv_A^2}{R} = 11mg$$

$$v_C = \sqrt{6gR}$$

$$N_C + mg = \frac{mv_C^2}{R} \text{ (Now solve and get)}$$

- Q.** If a ball is slightly displaced to right
Find the angle at which ball loses contact with the surface.

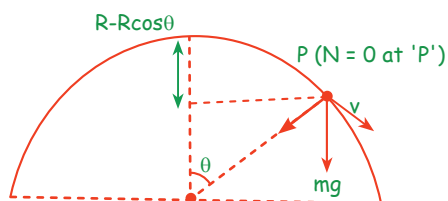


- Sol.** Suppose at point 'P' contact is lost ($N = 0$)

$$W_g + W_N = \Delta KE$$

$$mg(R - R\cos\theta) + 0 = \frac{1}{2}mv^2 - 0 \quad \dots(i)$$

$$\text{and } mg \cos\theta - N = \frac{mv^2}{R} \quad \dots(ii)$$



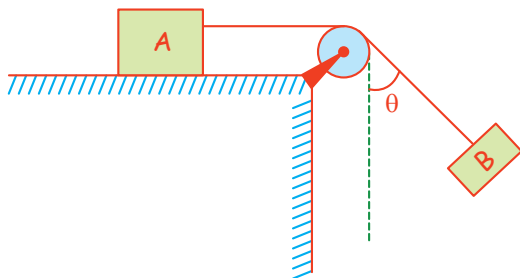
Solve (i) and (ii) and get

$$\cos\theta = 2/3$$

Abe kya tumne ye socha hai ki ye particle ground par kahan jaake girega bole to distance of the point from centre where it will strike the ground..... Chalo isko solve karo mast sawal hai and its ans. is 1.46 R.



- Q.** Two particles A and B each of mass m are connected by a massless string. A is placed on the rough table. The string passes over a small, smooth peg. B is left from a position making an angle θ with the vertical. Find the minimum coefficient of friction between A and the table so that A does not slip during the motion of mass B.



- Sol.** Block B rotates in vertical plane. Tension is maximum in string at lowest position. When block B is at lowest position and block A does not slide then it will not slide at any other position of B.

At lowest position

$$T - mg = \frac{mv^2}{\ell} \Rightarrow T = mg + \frac{mv^2}{\ell} \quad \dots (i)$$

From energy conservation

$$mg \ell (1 - \cos\theta) = \frac{1}{2} mv^2 \quad \dots (ii)$$

from equation (i) and (ii)

$$T = mg + 2mg(1 - \cos\theta) = 3mg - 2mg \cos\theta$$

For no slipping.

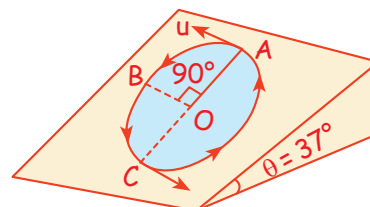
$$T = \mu mg$$

$$\Rightarrow 3mg - 2mg \cos\theta \leq \mu mg$$

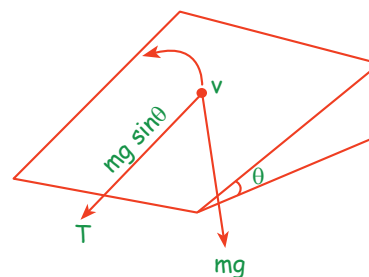
$$\Rightarrow \mu \geq 3 - 2 \cos\theta$$

$$\text{or } \mu_{\min} = 3 - 2 \cos\theta$$

- Q.** A small sphere of mass m is connected by a string to a nail at O and moves in a circle of radius ' r ' on the smooth plane inclined at an angle θ with the horizontal. If the sphere has a velocity u at the top position A. Mark the correct option(s).



- Sol.** (a)



$$T + mg \sin 37^\circ = \frac{mv^2}{r} \quad [\because T = 0]$$

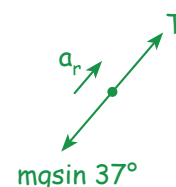
$$\therefore v = \sqrt{\frac{3gr}{5}}$$

$$(b) \quad mg(2r \sin 37^\circ) = \frac{1}{2} mv^2 - \frac{1}{2} m \left(\frac{3gr}{5} \right)$$

$$\therefore \frac{mv^2}{r} = 3mg$$

$$T - mg \sin 37^\circ = ma_r$$

$$\therefore T = \frac{18mg}{5}$$



man lo agar inclined plane rough hota $\mu = \tan \theta$ to pta karo tension kaha minimum hogi

at highest point

and it's wrong



Arey friction bhi toh hai

ye bhi meri galti hai



CONSERVATIVE FORCE

- ★ Those forces whose work done is independent of path taken i.e., their work depend on initial & final position only.
- ★ (WD) by such a force in a closed loop is always zero
- ★ Gravitation force, spring force, electrostatic force are the examples of conservative forces.
- ★ The concept of potential energy exists only in the case of conservative forces.

NON-CONSERVATIVE FORCE

- ★ A force is said to be non-conservative if work done by the force in moving a body depends upon the path taken between the initial and final positions.
- ★ Friction force, velocity-dependent forces such as air resistance, viscous force etc., are non-conservative forces.

POTENTIAL ENERGY(U)

Potential energy of the system is defined as negative of WD by internal conservative forces

$$\star \Delta U = -(\text{WD})_{\text{internal conservative forces (icf)}}$$

$$\star \Delta U = -\int \vec{F} \cdot d\vec{r}$$

$$\star dU = -\vec{F} \cdot d\vec{r}$$

$$\star \Delta U = U_f - U_i = -\text{WD} = -\int_i^f \vec{F} \cdot d\vec{r}$$

$$\star \Delta U \rightarrow \text{same for all frame}$$

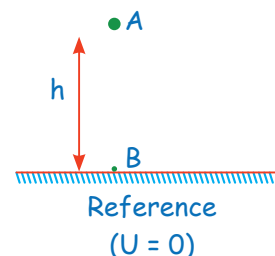
$$\star \text{P.E only defined for conservative force}$$

$$\star \text{P.E is defined for system of particle}$$

$$\star \text{P.E at a point depends on reference line}$$

But ΔU ("change" in P.E) it doesn't depend on Reference line

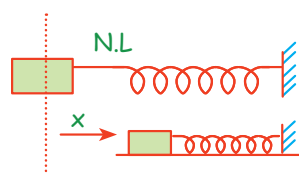
Gravitational Potential Energy



$$\Delta U = U_A - U_B = mgh$$

$$\text{If } U_B = 0 \Rightarrow U_A = mgh$$

Potential Energy in Spring



$$\Delta U = -(\text{WD})_{\text{icf}} = -(\text{WD})_{\text{sp}}$$

$$\Delta U = -\left(-\frac{1}{2}\right)k(x_f^2 - x_i^2)$$

$$U_f - U_i = \left(\frac{1}{2}\right)kx_f^2$$

$$\text{Since at N.L, } U_i = 0$$

$$\text{so } U_f = \left(\frac{1}{2}\right)kx_f^2$$

SKC



'h' upar P.E mgh tabhi hoti jab zameen par P.E = 0 manoge agar spring x daba hai ya x khincha to uske pet mein $\frac{1}{2}kx^2$ PE hogi.



Ab agle article mein maths maths and maths hai so sachin sir ka naam lo aur maths ka mind on karlo.

RELATION BETWEEN CONSERVATIVE FORCE & P.E

Jab F dekar U puche

$$dU = -\vec{F} \cdot d\vec{r}$$

$$\text{Let } \vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$$

$$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

$$\vec{F} \cdot d\vec{r} = F_x dx + F_y dy + F_z dz$$

$$\Delta U = U_f - U_i = -\int_i^f \vec{F} \cdot d\vec{r}$$

$$\Delta U = -[F_x dx + F_y dy + F_z dz]$$

Q. For a conservative force, $\vec{F} = -4x\hat{i} + 10\hat{j} + 20\hat{k}$
If P.E. at origin is 20J. Find P.E. at (1, 2, 3) will be?

$$\text{Sol. } U_f - 20 = -\left[\int_0^1 -4x dx + \int_0^2 10 dy + \int_0^3 20 dz\right]$$

Jab U dekar $\vec{F}$ puchhe tab:

$$\vec{F} = -\left[\frac{\partial U}{\partial x}\hat{i} + \frac{\partial U}{\partial y}\hat{j} + \frac{\partial U}{\partial z}\hat{k}\right]$$

$$F_x = -\frac{\partial U}{\partial x}; F_y = -\frac{\partial U}{\partial y}; F_z = -\frac{\partial U}{\partial z}$$

$\frac{\partial U}{\partial x}$ = differentiation of U wrt 'x' by keeping 'y' and 'z' constants. (Partial Differentiation)

$\frac{\partial U}{\partial y}$ = differentiation of U wrt 'y' by keeping 'x' and 'z' constant

$\frac{\partial U}{\partial z}$ = differentiation of U wrt 'z' by keeping 'x' and 'y' constant

Q. Given $U = x^2 y^2$ Find $\vec{F}$ at (2, 3).

$$\text{Sol. } F_x = -\frac{dU}{dx} = -2xy^2$$

Yahan definition ka ek - (minus) aa raha hai ise galti se bhi mat bhoolna warna pitaai hogi.



Yahan bhi definition ka ek - (minus) aa raha hai ise galti se bhi mat bhoolna warna pitaai hogi

$$\vec{F}_y = -\frac{dU}{dy} = -x^2 2y; F_z = -\frac{\partial U}{\partial z} = 0$$

$$\vec{F}_{\text{net}} = -2xy^2\hat{i} - x^2 2y\hat{j}$$

$$\text{at } (2, 3) \vec{F} = -36\hat{i} - 24\hat{j}$$

Q. If $U = 5xy$ Find $\vec{F}$ on particle at (2, 3)

$$\text{Sol. } F_x = -[5y]$$

$$F_y = -[5x], F_z = 0$$

$$\vec{F} = 15\hat{i} - 10\hat{j}$$

Q. Given $U = ax^2 - bxy^2$ solve for $\vec{F}$

$$\text{Sol. } F_x = -(a2x - by^2)$$

$$F_y = 0 - bx \cdot 2y \Rightarrow \vec{F} = (-2ax + by^2)\hat{i} + (2bxy)\hat{j}$$

Q. Given $U = x^2 + y^2 + z^2$, find $\vec{F}$.

$$\text{Sol. } F_x = -(2x + 0 + 0)$$

$$F_y = -(0 + 2y + 0)$$

$$F_z = -(0 + 0 + 2z) \Rightarrow \vec{F} = -2x\hat{i} - 2y\hat{j} - 2z\hat{k}$$

Q. Given $U = x^2 - 8x$ Find equilibrium position.

$$\text{Sol. at equilibrium } F = -\frac{\partial U}{\partial x} = 0$$

$$\text{or } \frac{dU}{dx} = 0$$

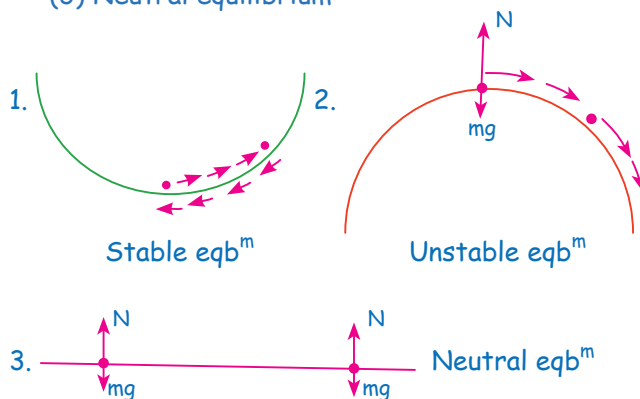
$$\Rightarrow 2x - 8 = 0$$

$$x = 4$$

TYPES OF EQUILIBRIUM

★ Type of equilibrium are

- (1) stable equilibrium
- (2) Unstable equilibrium
- (3) Neutral equilibrium



SKC

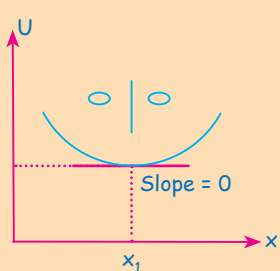
Dekh bhai result yeh hai ki, equilibrium position se thoda sa displace kar agar chhod kar gayi to banda unstable hai, agar wapas aayi to banda stable hai aur jab wapas aati hai to khushi (happy) hoti hai aur chhod kar jaati hai to dukh (sad) hota hai.



Kaam ka Dabba

★ $F = -\frac{dU}{dx} = -(\text{slope})$

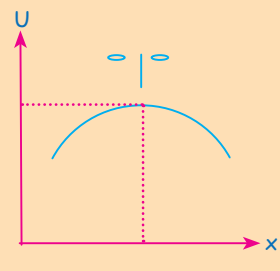
★ $F = -(\text{slope of } U \times \text{graph})$



At stable equil

slop = 0, $\frac{dU}{dx} = 0$

$\frac{d^2U}{dx^2} > 0$

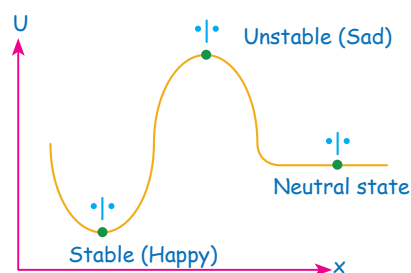


At unstable equilibm

slop = 0, $\frac{dU}{dx} = 0$

$\frac{d^2U}{dx^2} < 0$

Stable (Happy)	Unstable (Sad)	Neutral
$\frac{dU}{dx} = 0$	$\frac{dU}{dx} = 0$	$\frac{dU}{dx} = 0$
$\frac{d^2U}{dx^2} > 0$	$\frac{d^2U}{dx^2} < 0$	$\frac{d^2U}{dx^2} = 0$



Jab U ka x ki term mein function given ho aur stable or unstable pata karna ho sabse pehle $\frac{dU}{dx} = 0$ karke equilibrium position nikalo phir x ki un value par $\frac{d^2U}{dx^2}$ nikalo agar $\frac{d^2U}{dx^2} > 0$ aaya to stable equil aur $\frac{d^2U}{dx^2} < 0$ aaya to unstable equil aur $\frac{d^2U}{dx^2} = 0$ aaya to neutral equil.

Q. For a given potential $U = x^2 - 8x$, Find position of equilibrium and its type?

Sol. $\frac{dU}{dx} = 2x - 8 = 0 \Rightarrow x = 4$

Now $\frac{d^2U}{dx^2} = 2$ (ye to positive aa gya) $\frac{d^2U}{dx^2} > 0$

Hence, $x = 4$, is stable equilibrium point

Q. For a given potential $U = \frac{x^3}{3} - 9x$ find equilibrium and its types?

Sol. $\frac{dU}{dx} = \frac{3x^2}{3} - 9 = x^2 - 9$

Put $\frac{dU}{dx} = 0 \Rightarrow x^2 - 9 = 0$

$\Rightarrow x = +3, -3$ (equilibrium position)

Now $\frac{d^2U}{dx^2} = 2x$

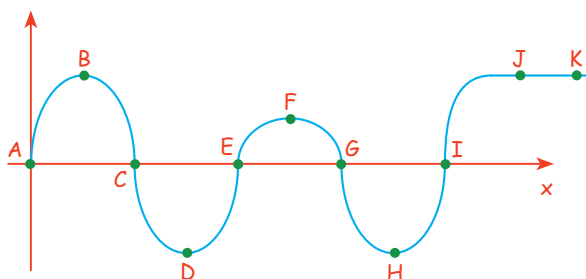
For $x = 3$ find $\frac{d^2U}{dx^2}$

$\frac{d^2U}{dx^2} = +6 > 0$ so, $x = 3$ is unstable equilibrium point)

For $x = -3$

$\frac{d^2U}{dx^2} = 2 \times -3 = -6$ so $x = -3$ is unstable equilibrium point)

Q. Identify stable & unstable equilibrium positions in following.



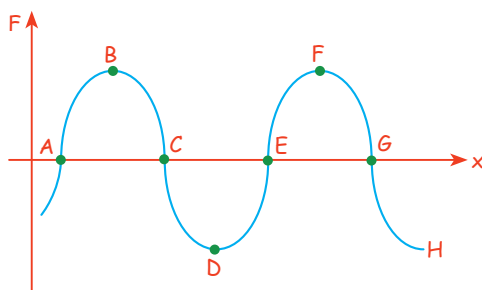
Sol. Slope = 0, $\frac{dU}{dx} = 0$ (at equilibrium)

B, F $\rightarrow$ Unstable equilibrium.

D, H $\rightarrow$ stable equilibrium

J, K: Neutral equilibrium

Q. In following figure, force(F) vs. Position 'x' curve is shown, Identify stable, unstable and neutral points



Sol. eqbm. vha hogi jaha $F = \frac{dU}{dx} = 0$;

So, A, C, E, G are equilibrium points.

A, E $\rightarrow$ stable $\left(\because \frac{dF}{dx} > 0 \right)$

C, G $\rightarrow$ unstable $\left(\because \frac{dF}{dx} < 0 \right)$

Q. The potential energy between two atoms in a molecule is given by, $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$, where 'a' and b are positive constants and x is the distance between the atoms. The system is in stable equilibrium when

Sol. Given that, $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$

$$\text{We, know } F = -\frac{dU}{dx} = -\left(\frac{-12a}{x^{13}} + \frac{6b}{x^7} \right) = 0$$

$$\text{or } \frac{6b}{x^7} = \frac{12a}{x^{13}} \text{ or } x^6 = 12a/6b = 2a/b$$

$$\text{or } x = \left(\frac{2a}{b} \right)^{1/6}$$

Q. The potential energy of a conservative system is given by $U = ax^2 - bx$ where a and b are positive constants. Find the equilibrium position and discuss whether the equilibrium is stable, unstable or neutral.

Sol. In a conservative field, $F = -\frac{dU}{dx}$

$$\therefore F = -\frac{d}{dx}(ax^2 - bx) = b - 2ax$$

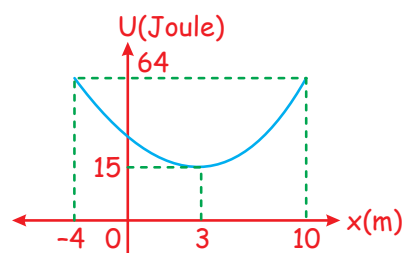
$$\text{For equilibrium, } F = 0 \text{ or } b - 2ax = 0 \Rightarrow x = \frac{b}{2a}$$

From the given equation we can see that

$$\frac{d^2U}{dx^2} = 2a \text{ (positive), i.e., } U \text{ is minimum.}$$

Therefore, $x = b/2a$ is the stable equilibrium position.

Q. A single conservative force $F(x)$ acts on a particle that moves along the x-axis. The graph of the potential energy with x is given. At $x = 5$ m, the particle has a kinetic energy of 50 J and its potential energy is related to position 'x' as $U = 15 + (x - 3)^2$ Joule, where x is in meter.



Sol. At $x = 5$ m, $KE = 50$ J

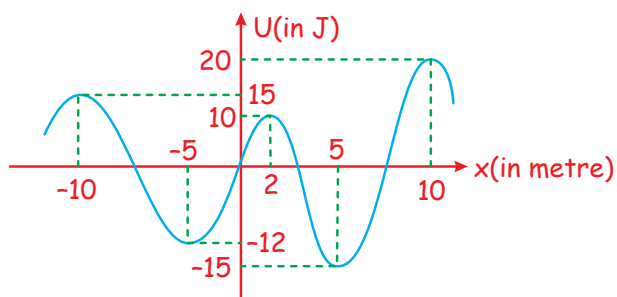
$$PE = 15 + (5 - 3)^2 = 19 \text{ J}$$

$$\text{Mechanical energy} = 69 \text{ J}$$

$$KE_{\max} = \text{Total Energy} - PE_{\min} = 69 - 15 = 54 \text{ J}$$

Comprehension questions:

In the figure the variation of potential energy of a particle of mass $m = 2$ kg is represented with respect to its x-coordinate. The particle moves under the effect of this conservative force along the x-axis.



(a) If it is released at the origin it will move in which direction?

Sol. It will move along the direction of force, as $\frac{dU}{dx} > 0$; $F = -\frac{dU}{dx}$ negative hence particle will move along negative x-axis towards origin.

(b) If the particle is released at $x = 2 + \Delta x$ where $\Delta x \rightarrow 0$ (it is positive) then its maximum speed in subsequent motion will be

Sol. When the particle is released at $x = 2 + \Delta x$ it will reach the point of least possible potential energy (-15 J) where it will have maximum kinetic energy.

$$\therefore \frac{1}{2}mv_{\max}^2 = 25 \Rightarrow v_{\max} = 5 \text{ m/s}$$

(c) $x = -5 \text{ m}$ and $x = 10 \text{ m}$ positions represents which type of equilibrium?

Sol. Conditions for stable equilibrium is:

$$\frac{dU}{dx} = 0 \text{ and } \frac{d^2U}{dx^2} > 0$$

Condition for unstable equilibrium is:

$$\frac{dU}{dx} = 0 \text{ and } \frac{d^2U}{dx^2} < 0$$

hence $x = -5$ is stable and $x = 10 \text{ m}$ is unstable equilibrium position

MECHANICAL ENERGY CONSERVATION

w_{iCF} = work by internal conservative force

According to work energy theorem

$$(w)_{\text{all the forces}} = \Delta KE$$

$$(w)_{\text{ext}} + (WD)_{iCF} + (WD)_{\text{internal non conservative}} = \Delta KE$$

If $w_{\text{ext}} = 0$ and $(WD)_{\text{internal non conservative}} = 0$

$$\Rightarrow 0 + (WD)_{iCF} + 0 = \Delta KE$$

$$-\Delta U = \Delta KE$$

$$-(U_f - U_i) = K_f - K_i$$

$$K_i + U_i = K_f + U_f$$

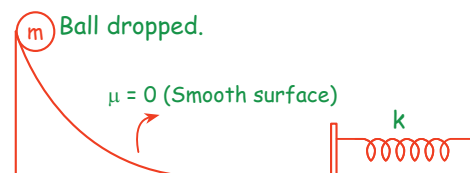
$$\text{or } (ME)_i = (ME)_f$$

(mechanical energy conservation)

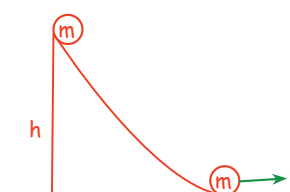
Agar hum dhyani se dekhe to ye WET ka ek special case hai jahan agar $(WD)_{\text{ext}} = 0$ and $(WD)_{\text{inc}} = 0$ hua to mechanical energy will conserve.



Q. Find maximum compression.



Sol.

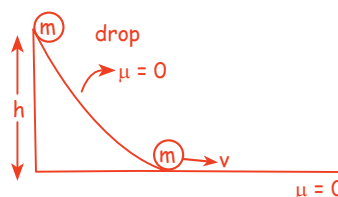


$$K_i + U_i = K_f + U_f$$

$$0 + mgh = 0 + \left(0 + \frac{1}{2}kx_{\max}^2\right)$$

$$x_{\max} = \sqrt{\frac{2mgh}{k}}$$

Q. Find final velocity of ball in following problem.



Sol. $W_g + W_N + W_f = \Delta KE$

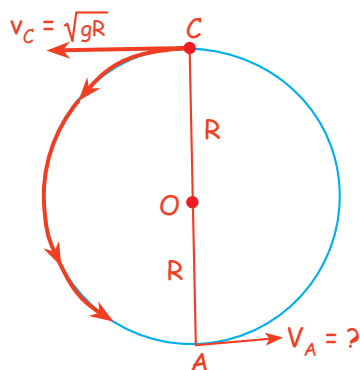
$$mgh + 0 + 0 = \frac{1}{2}mv^2 - 0, V = \sqrt{2gh}$$

By M.E. conservation

$$K_i + U_i = K_f + U_f$$

$$0 + mgh = \frac{1}{2}mv^2 + 0, V = \sqrt{2gh}$$

Q. Find velocity at point 'A' (V_A) if a particle performing vertical circular motion and starts at point 'c' with velocity $V_c = \sqrt{gR}$



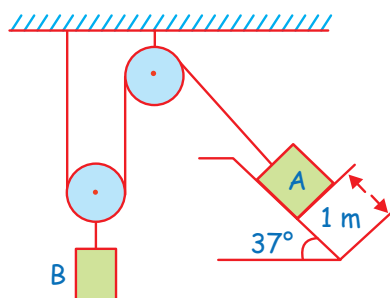
Sol. $U_C + K_C = U_A + K_A$

$$mg2R + \frac{1}{2} m (\sqrt{gh})^2 = 0 + \frac{1}{2} m V_A^2$$

$$2mgR + \frac{mgR}{2} = \frac{1}{2} m V_A^2$$

$$V_A = \sqrt{5gR}$$

Q. The blocks shown in figure have masses $M_A = 5$ kg and $M_B = 4$ kg. The system is released from rest. The speed of B after A has travelled a distance 1 m along the incline is



Sol. If A moves down the incline by 1 m, B shall move up by $\frac{1}{2}$ m. If the speed of B is v , then the speed of A will be $2v$.

From conservation of energy,

Gain in KE = loss in PE

$$\frac{1}{2} m_A (2v)^2 + \frac{1}{2} m_B v^2 = m_A g \times \frac{3}{5} - m_B g \times \frac{1}{2}$$

Solving, we get

$$v = \frac{1}{2} \sqrt{\frac{g}{3}}$$

POWER

1 Rate of work done

$$2 P = \frac{dw}{dt} = \frac{\vec{F} \cdot d\vec{r}}{dt}$$

Unit: Joule/sec. or Watt.

$$3 P = \vec{F} \cdot \vec{v}$$

$$4 P = \frac{dw}{dt} \text{ or } \int dw = \int P dt$$

$$5 WD = \int P dt$$

means P vs t ke graph ka area W.D dega

$$6 \text{ Average power} = \frac{\int P dt}{\int dt}$$

ye sare point
achhe se dekh lena



Q. If work as a function of 't' given as $w = 4t^2 + 3t$.
Find 'P' at $t = 2$ sec.

$$\text{Sol. } P = \frac{dw}{dt} = 8t + 3$$

$$\text{at } t = 2, P = 19 \text{ watt}$$

Q. Find power by friction & gravity at $t = 5$ sec.



$$\text{Sol. } f = 0.2 \times 100 = 20 \text{ N}$$

$$a = \mu g = 0.2 \times 10 = 2 \text{ (deceleration)}$$

$$\text{Velocity at } t = 5 \text{ will be, } v = 100 - 2 \times 5 = 90 \text{ m/s}$$

Power by friction at $t = 5$ sec

$$= \vec{F} \cdot \vec{v} = f v \cos 180^\circ = -20 \times 90 = -1800 \text{ watt}$$

$$\text{Power supply by gravity at } t = 5 \text{ sec} = \vec{F}_g \cdot \vec{v} = (mg) \cdot v \cos 90^\circ = 0,$$

$$P = 0 \quad (\text{Bcz } \theta = 90^\circ)$$

Q. $P = 2t^2 + t$ Find power at $t = 2$ sec

$$\text{Sol. } P_{\text{inst}} = 2 \times 2^2 + 2 = 10$$

Find avg. power from $t = 0 \rightarrow t = 2$ sec

$$\text{Avg. power} = \frac{\int P dt}{\int dt} = \int_0^2 \frac{(2t^2 + t) dt}{dt} \text{ (solve)}$$

Q. A particle of mass 2 kg is projected with speed 100 m/s at an angle 37° with horizontal. Find power supply by gravity at $t = 2$ sec.

Sol. At $t = 2$ $\vec{v} = 80\hat{i} + 40\hat{j}$ and $\vec{F} = mg = -20\hat{j}$
 $P = \vec{F} \cdot \vec{v} = -20 \times 40 = -800$ watt



ye question har saal
JEE Mains mein
pucha ja rha hai iska
result yaad krlo

Q. A body is moving unidirectionally under the influence of a constant power source. Its displacement in time t is proportional to:

Sol. $P = Fv = \text{constant}$

$$\Rightarrow m \frac{dv}{dt} v = \text{constant}$$

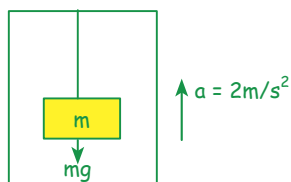
$$\int_0^v v dv = (C) \int_0^t dt \Rightarrow \left(\frac{v^2}{2} \right) = Ct$$

$$v \propto t^{1/2} \Rightarrow \frac{ds}{dt} \propto t^{1/2}$$

$$\int_0^s ds = K \int_0^t t^{1/2} dt \Rightarrow S = K \times \frac{2}{3} t^{3/2}$$

$$S \propto t^{3/2}$$

Q. Lift starts moving upward from rest having acceleration 2m/s^2 upward at $t = 0$. Find Power supply by tension & gravity to the block at $t = 5$ sec in ground frame ($m = 2\text{kg}$).



Sol. $P = \vec{F} \cdot \vec{v}$

$$v = u + at = 0 + 2 \times 5 \quad v = 10\text{m/s}$$

$$\text{Power supplied by gravity} = -mgv = -20 \times 10 \\ = -200 \text{ watt}$$

$$\text{Power supplied by tension force} = +T.v \\ = 24 \times 10 = 240 \text{ watt}$$

$$\Rightarrow T = 20 + 2 \times 2 = 24 \text{ N } (\because T = T_g + ma)$$



Kaam ka Dabba

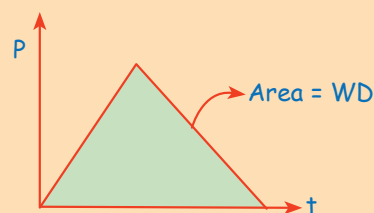
$$\star P = \vec{F} \cdot \vec{v}$$

$$\star P = \frac{dw}{dt} \text{ (instantaneous power)}$$

$$\star WD = \int P dt$$

$$\star \text{Avrg power} = \frac{\int p dt}{\int dt}$$

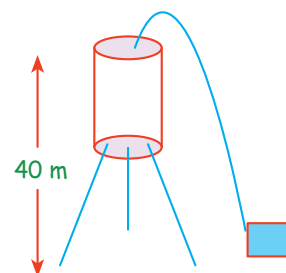
$\star$ Area of power time graph given work.



$$WD = \int P dt = \text{area under curve in } P \text{ vs } t \text{ graph}$$

$$\star \text{Efficiency} = \frac{\text{output power}}{\text{input power}}$$

Q. A water pump rated 600 watt has efficiency 75%. If it is employ to raise water through a height of 40 meter. Find the mass of water drawn in 2 hour.



$$\text{Sol. } \frac{dw}{dt} = \frac{d}{dt}(mgh)$$

$$P = \frac{dw}{dt} = gh \left(\frac{dm}{dt} \right)$$

$$\boxed{\frac{dm}{dt} = \frac{p}{gh}}$$

$$\text{Given } P = 600 \times \frac{75}{100} = 450 \text{ watt}$$

$$\text{water drawn in 2 hour} = \frac{450}{10 \times 40} \times (3600 \times 2)$$

$$= 8100 \text{ kg} = 8.1 \text{ L}$$

Q. Wind enters in a wind mill with a velocity of 20 m/sec. Facing area of the windmill is 10 m^2 and density of air is 1.2 kg/m^3 . If wind energy is converted into electrical energy with 33.3% efficiency, then electrical power produced by the wind mill in kW is

Sol. Energy entering in the windmill = $\frac{1}{2}mv^2$

$$P_{\text{in}} = \frac{dE}{dt} = \left(\frac{1}{2}v^2 \right) \left(\frac{dm}{dt} \right)$$

$$P_{\text{in}} = \left(\frac{1}{2}v^2 \right) (\rho Av) = \frac{1}{2} \rho Av^3$$

Electrical power output

$$P_{\text{out}} = \frac{1}{3} \left(\frac{1}{2} \rho Av^3 \right)$$

$$P_{\text{out}} = \frac{1}{6} \rho Av^3 = \frac{1}{6} \times 1.2 \times 10 \times (20)^3$$

$$\Rightarrow P_{\text{out}} = 16 \text{ kW}$$



Kaam ka Dabba

Let us summarise the concepts developed so far in this chapter.

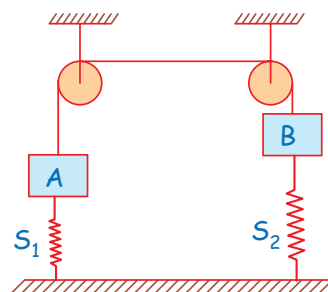
- ★ Work done on a particle is equal to the change in its kinetic energy.
- ★ Work done on a system by all the (external and internal) forces is equal to the change in its kinetic energy.
- ★ A force is called conservative if the work done by it during a round trip of a system is always zero. The force of gravitation, Coulomb force, force by a spring etc. are conservative. If the work done by it during a round trip is not zero, the force is nonconservative. Friction is an example of nonconservative force.
- ★ The change in the potential energy of a system corresponding to conservative internal forces is equal to negative of the work done by these forces.

- ★ If no external forces act (or the work done by them is zero) and the internal forces are conservative, the mechanical energy of the system remains constant. This is known as the principle of conservation of mechanical energy.
- ★ If some of the internal forces are nonconservative, the mechanical energy of the system is not constant.
- ★ If the internal forces are conservative, the work done by the external forces is equal to the change in mechanical energy.

Now Practice following question for Advance

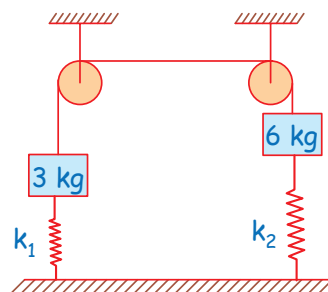


Q. In the figure shown below masses of blocks A and B are 3 kg and 6 kg respectively. The force constants of springs S_1 and S_2 are 160 N/m and 40 N/m respectively. Length of the light string connecting the blocks is 8 m. The system is released from rest with the springs at their natural lengths. The maximum elongation of spring S_1 will be:



Sol. $\Delta K = 0 \Rightarrow \Delta U_{\text{gravity}} = -6gx + 3gx$

$$\Delta U_{\text{spring}} = \frac{1}{2}k_2x^2 + \frac{1}{2}k_1x^2$$



Applying, $\Delta K + \Delta U_{\text{gravity}} + \Delta U_{\text{spring}} = 0$

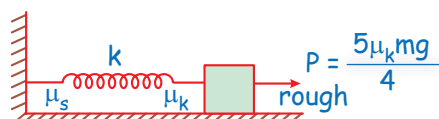
$$6g(x) - 3g(x) = \frac{1}{2}k_2x^2 + \frac{1}{2}k_1x^2 \quad [\because \Delta K = 0]$$

$$6g = (k_1 + k_2)x, x = \frac{6 \times 9.8}{200} = \frac{3 \times 9.8}{100} \Rightarrow x = 0.294 \text{ m}$$

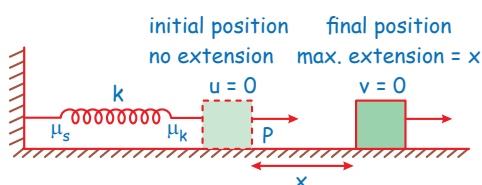
- Q.** A block of mass m rests on a rough horizontal plane having coefficient of kinetic friction μ_k . The spring is in its natural length, when a constant force of magnitude $P = \frac{5\mu_k mg}{4}$

starts acting on the block. The spring force F is a function of extension x varies as $F = kx^3$. (where k is spring constant)

Find the maximum extension in the spring (Assume that the force P is sufficient to make the block move).



- Sol.** For motion to start $\frac{5\mu_k mg}{4} > \mu_s mg$ or $5\mu_k > 4\mu_s$



At the final position of the block extension in spring is maximum and the speed of the block is $v = 0$. Hence the net work done in taking the block from initial to final position

$\Delta W = (\text{Work done by force } P + \text{work done by spring force } F + \text{work done by friction}) = \Delta K = 0$

$$= Px - \int_0^x Kx^3 \cdot dx - \mu_k mgx$$

$$= \frac{5\mu_k mg}{4} \cdot x - \frac{kx^4}{4} - \mu_k mgx$$

Solving we get, $x = \left(\frac{\mu_k mg}{k} \right)^{1/3}$

- Q.** A particle of mass m is initially at rest at the origin. It is subjected to a force and starts moving along the x -axis. Its kinetic energy K changes with time as $dK/dt = \gamma t$ where γ is a positive constant of appropriate dimensions. Which of the following statements is (are) true?

Sol. $\frac{dK}{dt} = mv \frac{dv}{dt} = \gamma t$

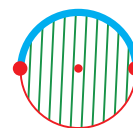
$$v dv = \frac{\gamma}{m} t dt$$

$$\frac{v^2}{2} = \frac{\gamma}{m} \frac{t^2}{2} \Rightarrow v \propto t$$

$$\frac{dv}{dt} = \text{constant} \Rightarrow F = \text{constant}$$

$$\frac{dx}{dt} \propto t \Rightarrow x \propto t^2$$

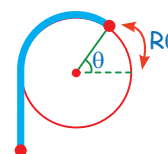
- Q.** Two identical particles are attached by an inextensible and massless string which passes over a smooth fixed cylinder of radius ' R ' as shown in figure. Initially particles lie on same horizontal line and they are slightly displaced. The velocity of particles as a function of θ is given by



- Sol.** Loss in P.E. = gain in K.E.

$$(mgR\theta - mgR\sin\theta) = \frac{1}{2} \times 2m \times v^2$$

Solve for ' v '



- Q.** The angle θ at which the particle slipping on the cylinder leaves contact with the cylinder, satisfy the relation.

- Sol.** At the breaking point $N = 0$

$$\frac{mv^2}{R} = mg \cos \left(\theta - \frac{\pi}{2} \right)$$

$$v^2 = gR \sin\theta$$

$$gR(\theta - \sin\theta) = gR \sin\theta$$

$$\sin\theta = \theta/2$$

- Q.** The tangential acceleration of particle in contact with cylinder is

Sol. $T + mg\sin(\theta - \pi/2) = ma$ (i)

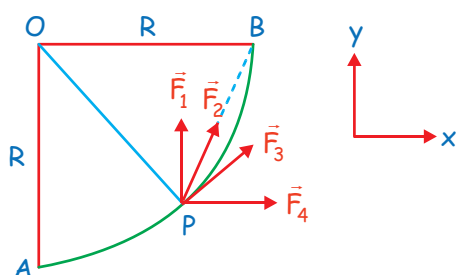
$$mg - T = ma$$
(ii)

on (i) + (ii)

$$mg(1 - \cos\theta) = 2ma$$

$$a = g \sin^2(\theta/2)$$

- Q. AB is a quarter of a smooth horizontal circular track of radius R. A particle P of mass m moves along the track from A to B under the action of following forces:



$$\vec{F}_1 = F \text{ (always towards y-axis)}$$

$$\vec{F}_2 = F \text{ (always towards point B)}$$

$$\vec{F}_3 = F \text{ (always along the tangent to path AB)}$$

$$\vec{F}_4 = F \text{ (always towards x-axis)}$$

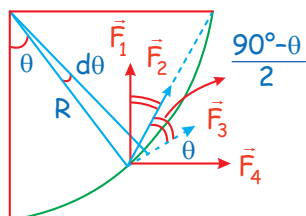
Column-I		Column-II	
A.	Work done by $\vec{F}_1$	p.	$\sqrt{2}FR$
B.	Work done by $\vec{F}_2$	q.	$\frac{1}{\sqrt{2}}FR$
C.	Work done by $\vec{F}_3$	r.	FR
D.	Work done by $\vec{F}_4$	s.	$\frac{\pi FR}{2}$

Sol. A-(r); B-(p); C-(s); D-(r)

For (A): Work done $\vec{F}_1 = FR$

For (B):

$$dW = \vec{F} \cdot d\vec{s} = (FRd\theta)\cos$$



$$W = \int_0^{\pi/4} FR \cos\left(45^\circ - \frac{\theta}{2}\right)$$

$$\text{For (C): } W = \int \vec{F} \cdot d\vec{s} = F\left(\frac{\pi R}{2}\right) = \frac{\pi FR}{2}$$

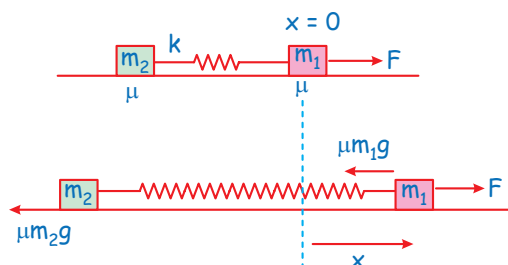
$$\text{For (D): } W = \int \vec{F} \cdot d\vec{s} = (F)(R) = FR$$

- Q. Two blocks of masses m_1 and m_2 are connected by a spring of stiffness $k = 200 \text{ Nm}^{-1}$. The coefficient of friction between the blocks and the surface is μ . Find the minimum constant horizontal force F (in Newton) to be applied to m_1 in order to slide the mass m_2 . (Initially spring is in its natural length).

(Take $m_1 = 3 \text{ kg}$, $m_2 = 5 \text{ kg}$, $g = 10 \text{ m/s}^2$, $\mu = 0.2$)



Sol.



$$W_F + W_{Sp} + W_{fric} = \Delta k$$

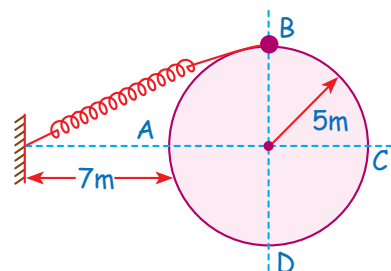
$$\Rightarrow Fx - \frac{1}{2} kx^2 - \mu m_1 g x = 0$$

$$\text{and } kx = \mu m_2 g$$

$$\Rightarrow F - \frac{1}{2} \mu m_2 g - \mu m_1 g = 0 \Rightarrow F = \mu m_1 g + \frac{\mu m_2 g}{2}$$

- Q. A collar B of mass 2kg is constrained to move along a vertical smooth and fixed circular track of radius 5m as shown in figure. The spring is in plane of the track and has a spring constant of 200 N/m. It is underformed when collar is at A. It starts from rest at B. What is the normal force (in N) exerted by the track on the collar when it passes through A?

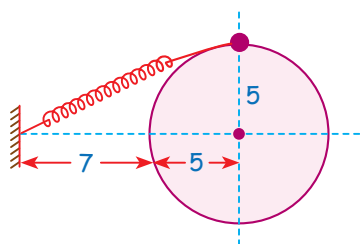
[$g = 10 \text{ m/s}^2$]



$$\text{Sol. } l_0 + x = \sqrt{5^2 + 12^2} = 13$$

$$x = 6\text{m}$$

$$mg \times 5 + \frac{1}{2} k (6^2 - 0^2) = \frac{1}{2} mv^2$$

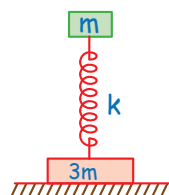


$$100 + \frac{1}{2} \times 200 \times 36 = \frac{1}{2} \times 2 \times v^2$$

$$v^2 = 3700$$

$$N = \frac{mv^2}{R} = \frac{2 \times 3700}{5} = 1480 \text{ N}$$

- Q.** In the figure shown the spring constant is k . The mass of the upper disc is m and that of the lower disc is $3m$. The upper block is depressed down from its equilibrium position by a distance $\delta = 5mg/k$ and released at $t = 0$. Find the velocity of ' m ' when normal reaction on $3m$ is mg .



Sol. $N = mg$

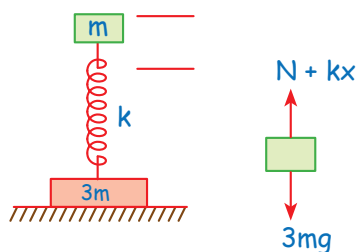
$$N + kx = 3mg$$

$$mg + kx = 3mg$$

$$x = \frac{2mg}{k}$$

Applying C.O.E.

$$\frac{1}{2} k \left(\frac{6mg}{k} \right)^2 - \frac{1}{2} k \left(\frac{6mg}{k} \right)^2 - \frac{1}{2} k \left(\frac{2mg}{k} \right)^2 - \frac{8mg}{k} mg = \frac{1}{2} mv^2$$



$$\frac{36m^2g^2}{k} - \frac{4m^2g^2}{k} - \frac{16m^2g^2}{k} = mv^2$$

$$\frac{16m^2g^2}{k} = mv^2 \Rightarrow v = 4g \left[\frac{m}{k} \right]^{\frac{1}{2}}$$

- Q.** A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration (a) is varying with time t as $a = k^2rt^2$, where k is a constant.

The power delivered to the particle by the force acting on it is given as

Sol. Centripetal acceleration, $a = k^2rt^2$

Since, centripetal acceleration is $\frac{v^2}{r}$

$$\therefore \frac{v^2}{r} = k^2rt^2$$

$$v = krt$$

Centripetal force is perpendicular to the tangential velocity, therefore power delivered by it will be zero.

Tangential acceleration,

$$\Rightarrow a_t = \frac{dv}{dt} = kr$$

$$\Rightarrow \text{Tangential force, } F_t = ma_t = mkr$$

$$\therefore \text{Power delivered, } P = F_tv = (mkr)(krt) = mk^2r^2t$$

- Q.** A particle moving in a circular track such that is gravitational potential energy $U = -\frac{k}{r}$. Find it's speed, kinetic energy, total energy.

Sol. $U = -\frac{k}{r}$

$$F = -\frac{\partial U}{\partial r} = -\frac{k}{r^2}$$

$$F = \frac{k}{r^2} = \frac{mv^2}{r} \text{ (magnitude)}$$

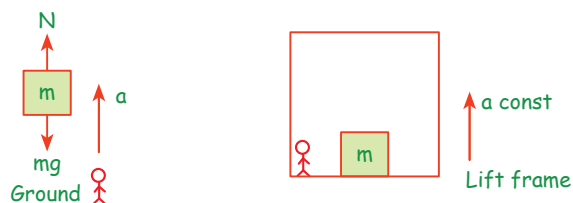
$$v = \sqrt{\frac{k}{mr}}$$

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}m \left(\frac{k}{mr} \right) = \frac{k}{2r}$$

$$TE = -\frac{k}{r} + \frac{k}{2r} = -\frac{k}{2r}$$

Q. A block of mass M is placed inside the lift which starts moving from rest to upward direction with constant acc. ' a '. Find WD by all the forces on the block in ground frame and in lift frame.

Sol.



$$N - mg = ma \Rightarrow N = m(g + a)$$

displacement of block in ' t_0 ' time.

$$h = y = 0 + \frac{1}{2}at_0^2$$

✦ (WD) by gravity in ground frame = $-mgh$

✦ (WD) by normal force in ground frame
 $= Nh = +Ny = m(g + a) \frac{1}{2}at_0^2$

✦ (WD) by pseudo force in ground frame = 0

✦ (WD) by gravity in lift frame
 $W = mg \times 0 = 0$

✦ (WD) by normal force in lift frame = 0

✦ (WD) by pseudo force in lift frame
 $= (ma) \times 0 = 0$

✦ That shows that WD by a force is also frame of reference dependent.